

- 1. RDSO trials report(NCR) DAY-1**
- 2. RDSO trials report(NCR) DAY-2**
- 3. Lab Trial Report (NCR)**

For

Train Collision Avoidance System

Prepared By
HBL Power Systems Limited
Hyderabad

#	Day1 - July 18 th Trial Details	
1	Date	18/07/24
2	Time	09:30 to 11:45 hrs
3	Loco ID	23545
4	Loco Configuration	Vande Bharat DTC 8 Coaches
5	Loco Type	Vande Bharat (HBL)
6	Division	NCR
7	Trial Section	RDE - VRBD (UP Direction)

1. Check SPAD prevention

Test Scenario	Input Specification	Expected Result	Actual Result
Ensure Loco TCAS is registered with the Sholaka Station TCAS and travelling in FS Mode at max section speed.	Railway staff performs the operation to keep home signal S75 to Red aspect.	Status of the Home signal S75 shall be displayed on LP-OCIP (and MA shall be displayed up to the Home signal)	1) This SPAD was conducted at Home signal S75 . 2) FSB was applied by LTCAS at a loco speed of 158 kmph when MA was 1516 mtrs to the Home signal S75 at ABS LOC 1463.425
	Run the Loco with maximum section speed.	Loco TCAS shall apply brakes and Loco shall stop before the Home signal.	3) Loco applied FSB upto speed reduction 33 kmph. Then FSB was released at AbsLoc 1462.288 KM and the Loco moved at speeds around 15 kmph for a few meters. 4) Loco re-applied FSB (at AbsLoc 1462.110) and halted at ABS LOC 1462.084 which is 7 mtrs before the Home signal S75.

Day 1 – RDSO trials report (NCR)

HBL®

Locold	Abs Location	Train Len	Train Speed	Dir	Emergency Mode	Tag-Id	TIN	BrakeStatus	StPrfld	StPrptoMA	HlthBits	MAC	CRC	Time stamp
23545	1463684	250	157	R	-	FS	191 95	-	3	1	00	5A62 4F8A	09:50:01	
23545	1463596	250	157	R	-	FS	191 95	-	3	1	00	8387 82E0	09:50:03	
23545	1463488	250	158	R	-	FS	193 95	-	3	1	00	4295 EB02	09:50:05	
23545	1463425	250	158	R	-	FS	193 95	FSB	3	1	00	0023 2817	09:50:07	
23545	1463337	250	158	R	-	FS	193 95	FSB	3	1	00	7834 C5A0	09:50:09	
23545	1463208	250	156	R	-	FS	195 95	FSB	3	1	00	2559 8CFA	09:50:11	
23545	1463139	250	149	R	-	FS	195 95	FSB	3	1	00	8A81 D11C	09:50:13	
23545	1463058	250	142	R	-	FS	195 95	FSB	3	1	00	0233 57E9	09:50:15	
23545	1462981	250	135	R	-	FS	195 95	FSB	3	1	00	0DF8 1D94	09:50:17	
23545	1462908	250	127	R	-	FS	195 95	FSB	3	1	00	C7B1 F7F7	09:50:19	
23545	1462839	250	121	R	-	FS	195 95	FSB	3	1	00	BF38 328C	09:50:21	
23545	1462773	250	114	R	-	FS	195 95	FSB	3	1	00	885D F8AE	09:50:23	
23545	1462712	250	107	R	-	FS	195 95	FSB	3	1	00	A7AE 017D	09:50:25	
23545	1462654	250	100	R	-	FS	195 95	FSB	3	1	00	1224 3153	09:50:27	
23545	1462601	250	94	R	-	FS	195 95	FSB	3	1	00	F355 9F2F	09:50:29	
23545	1462550	250	88	R	-	FS	195 95	FSB	3	1	00	6427 4AC0	09:50:31	
23545	1462503	250	81	R	-	FS	195 95	FSB	3	1	00	763C C0A5	09:50:33	
23545	1462460	250	75	R	-	FS	195 95	FSB	3	1	00	57E8 013D	09:50:35	
23545	1462420	250	67	R	-	FS	195 95	FSB	3	1	00	4F02 8044	09:50:37	
23545	1462385	250	61	R	-	FS	195 95	FSB	3	1	00	C804 3AE9	09:50:39	
23545	1462353	250	54	R	-	FS	195 95	FSB	3	1	00	8523 D5AA	09:50:41	
23545	1462328	250	47	R	-	FS	197 95	FSB	3	1	00	88C9 3EB3	09:50:43	
23545	1462308	250	41	R	-	FS	197 95	FSB	3	1	00	4906 1648	09:50:45	
23545	1462288	250	33	R	-	FS	197 95	FSB	3	1	00	BD1C 7DB7	09:50:47	
23545	1462272	250	25	R	-	FS	197 95	-	3	1	00	FF9A C7FD	09:50:49	
23545	1462260	250	18	R	-	FS	197 95	-	3	1	00	B1B7 EA9F	09:50:51	
23545	1462251	250	16	R	-	FS	197 95	-	3	1	00	DDF9 4CD8	09:50:53	

Locold	Abs Location	Train Len	Train Speed	Dir	Emergency	Mode	Tag-Id	TIN	BrakeStatus	StPrfld	StPrptoMA	HlthBits	MAC	CRC	Time stamp
23545	1462128	250	15	R	-	FS	199	95	-	3	1	00	147D	7D15	09:51:21
23545	1462119	250	16	R	-	FS	199	95	-	3	1	00	F43D	08F4	09:51:23
23545	1462110	250	16	R	-	FS	199	95	FSB	3	1	00	65D6	233D	09:51:25
23545	1462102	250	15	R	-	FS	199	95	FSB	3	1	00	A2F8	9ACC	09:51:27
23545	1462093	250	14	R	-	FS	199	95	FSB	3	1	00	2516	9682	09:51:29
23545	1462087	250	8	R	-	FS	199	95	FSB	3	1	00	D6EA	7F61	09:51:31
23545	1462084	250	2	R	-	FS	199	95	FSB	3	1	00	F923	F35C	09:51:33
23545	1462084	250	0	R	-	FS	199	95	-	3	1	00	466C	8952	09:51:35
23545	1462084	250	0	R	-	FS	199	95	-	3	1	00	AF0D	893C	09:51:37
23545	1462084	250	0	R	-	FS	199	95	-	3	1	00	E404	956B	09:51:39
23545	1462084	250	0	R	-	FS	199	95	-	3	1	00	8F2C	51A5	09:51:41
23545	1462084	250	0	R	-	FS	199	95	-	3	1	00	32C7	5684	09:51:43
23545	1462084	250	0	R	-	FS	199	95	-	3	1	00	99BC	9DAA	09:51:45

Figure: FSB applied to reduce speed and prevent SPAD



Figure: LP-OCIP showing the application of FSB



Figure: LP-OCIP showing Loco stopped 7 mtrs before the signal

2. Verification of loop line speed control

Test Scenario	Input Specification	Expected Result	Actual Result
Loco TCAS is registered with the Station TCAS and travelling in FS Mode and approaching towards Hodal loop line	Loco is travelling at maximum section speed on approach of Hodal (HDL) loop line in FS mode	At Hodal first loop line, the permissible speed is 30 kmph. Loco TCAS shall apply brakes to reduce the speed within the permissible speed (30 kmph) in the turnout when Loco is in FS mode, the same can be seen on LP-OCIP.	- Loco was travelling at a speed of 160 kmph in UP direction. - FSB was applied at a speed of 158 Kmph in UP direction @Abs Loc 1452.276 and the Loco speed controlled to 29 kmph on reaching the Turn Out point in FS mode @AbsLoc 1450.989 - After entering the loop line, the loco maintained the speed below 30kmph.

Locoid	Abs Location	Train Len	Train Speed	Dir	Emergency	Mode	Tag-Id	TIN	BrakeStatus	StPrfld	StPrptoMA	HlthBits	MAC	CRC	Time stamp
23545	1452542	250	159	R	-	FS	479	99	-	52	1	00	A00C	2001	09:56:45
23545	1452453	250	159	R	-	FS	479	99	-	52	1	00	F10B	F07B	09:56:47
23545	1452365	250	158	R	-	FS	479	99	-	52	1	00	0729	6F72	09:56:49
23545	1452276	250	158	R	-	FS	479	99	FSB	52	1	08	E31A	8F22	09:56:51
23545	1452179	250	158	R	-	FS	481	99	FSB	52	1	00	142F	1A7C	09:56:53
23545	1452092	250	154	R	-	FS	481	99	FSB	52	1	00	6A6E	C4FD	09:56:55
23545	1452008	250	148	R	-	FS	481	99	FSB	52	1	00	A685	81D2	09:56:57
23545	1451908	250	142	R	-	FS	611	99	FSB	52	1	08	6862	AA5C	09:56:59
23545	1451853	250	135	R	-	FS	611	99	FSB	52	1	00	1E91	B8A3	09:57:01
23545	1451782	250	129	R	-	FS	563	1	FSB	52	1	00	48AB	78EE	09:57:03
23545	1451712	250	122	R	-	FS	563	1	FSB	52	1	00	8007	AF52	09:57:05
23545	1451646	250	116	R	-	FS	563	1	FSB	52	1	08	60DE	1532	09:57:07
23545	1451583	250	110	R	-	FS	563	1	FSB	52	1	00	E685	E0DC	09:57:09
23545	1451524	250	103	R	-	FS	563	1	FSB	52	1	00	23B9	98B5	09:57:11
23545	1451468	250	97	R	-	FS	563	1	FSB	52	1	00	5D67	DF70	09:57:13
23545	1451415	250	92	R	-	FS	563	1	FSB	52	1	08	F1EF	4E84	09:57:15
23545	1451357	250	85	R	-	FS	565	1	FSB	52	1	00	5ED0	1423	09:57:17
23545	1451308	250	80	R	-	FS	567	3	FSB	52	1	00	FA6D	D1F4	09:57:19
23545	1451269	250	72	R	-	FS	567	3	FSB	52	1	00	69F9	580F	09:57:21
23545	1451231	250	65	R	-	FS	567	3	FSB	52	1	08	F558	21DD	09:57:23
23545	speed under 30 kmph	59	59	R	-	FS	567	3	FSB	52	1	00	E77F	884F	09:57:25
23545	1451188	250	52	R	-	FS	567	3	FSB	52	1	00	34B2	F9E6	09:57:27
23545	1451139	250	45	R	-	FS	567	3	FSB	52	1	00	81AF	35BE	09:57:29
23545	1451113	250	38	R	-	FS	569	5	-	52	1	08	6A8C	84E8	09:57:31
23545	1451094	250	30	R	-	FS	569	5	-	52	1	00	FF76	17F6	09:57:33
23545	1451077	250	30	R	-	FS	569	5	-	52	1	00	B4D5	80A8	09:57:35
23545	1451060	250	30	R	-	FS	569	5	-	52	1	00	7930	C129	09:57:37
23545	1451043	250	30	R	-	FS	569	5	-	52	1	08	9901	E10F	09:57:39
23545	1451027	250	29	R	-	FS	569	5	-	52	1	00	58C1	E6A1	09:57:41
23545	1451010	250	29	R	-	FS	569	5	-	52	1	00	20D5	D3A9	09:57:43
23545	1450994	250	29	R	-	FS	569	5	-	52	1	00	D91C	1FED	09:57:45
23545	1450977	250	29	R	-	FS	569	5	-	52	1	08	DE76	B5D7	09:57:47
23545	1450961	250	29	R	-	FS	569	5	-	52	1	00	38E1	1923	09:57:49
23545	1450944	250	29	R	-	FS	569	5	-	52	1	00	1C14	94D2	09:57:51
23545	1450928	250	29	R	-	FS	569	5	-	52	1	00	A2E6	E0B6	09:57:53

Figure: FSB applied and speed controlled to under 30 kmph before loopline

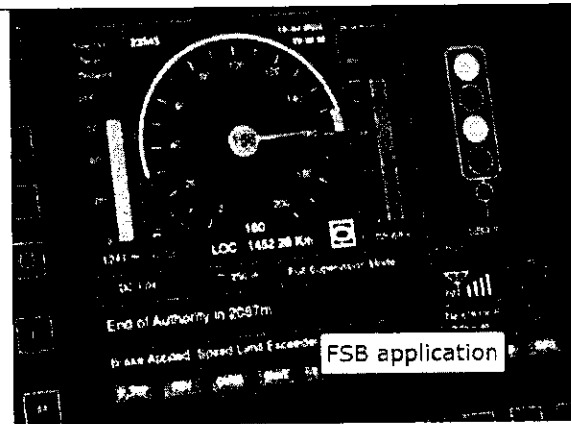


Figure: LP-OCIP showing the application of FSB



Figure: LP-OCIP showing the speed (under 30kmph)

3. Verification of loop line speed control

Test Scenario	Input Specification	Expected Result	Actual Result
Loco TCAS is registered with the Station TCAS and travelling in FS Mode and approaching towards Chhata loop line	Loco is travelling at maximum section speed on approach of Chhata (CHJ) loop line in FS mode	At Chhata first loop line, the permissible speed is 30 kmph. Loco TCAS shall apply brakes to reduce the speed within the permissible speed (30 kmph) in the turnout when Loco is in FS mode, the same can be seen on LP-OCIP.	- Loco was travelling at a speed of 160 kmph in UP direction. - FSB was applied at a speed of 155 Kmph in UP direction @Abs Loc 1429.574 and the Loco speed controlled to 30 kmph on reaching the Turn Out point in FS mode @AbsLoc 1428.320 - After entering the loop line, the loco maintained the speed below 30kmph.

Locoid	Abs Location	Train Len	Train Speed	Dir	Emergency	Mode	Tag-Id	TIN	BrakeStatus	StPrfid	StPrptoMA	HthBits	MAC	CRC	Time stamp
23545	1429745	250	156	R	-	FS	35	107	-	27	1	00	FD48	C982	10:18:39
23545	1429638	250	156	R	-	FS	37	107	-	27	1	00	F8C4	C128	10:18:41
23545	1429574	250	155	R	-	FS	37	107	FSB	27	1	08	2492	1C20	10:18:43
23545	1429488	250	154	R	-	FS	37	107	FSB	27	1	00	5D89	2DAF	10:18:45
23545	1429388	250	150	R	-	FS	121	107	FSB	27	1	00	62AC	B86D	10:18:47
23545	1429327	250	142	R	-	FS	171	107	FSB	27	1	00	594C	47ED	10:18:49
23545	1429250	250	135	R	-	FS	171	107	FSB	27	1	08	0030	D95C	10:18:51
23545	1429169	250	128	R	-	FS	123	1	FSB	27	1	00	0AA1	B189	10:18:53
23545	1429100	250	121	R	-	FS	123	1	FSB	27	1	00	F1DB	5707	10:18:55
23545	1429035	250	114	R	-	FS	123	1	FSB	27	1	00	D3C1	9F93	10:18:57
23545	1428974	250	107	R	-	FS	123	1	FSB	27	1	08	A64D	8894	10:18:59
23545	1428916	250	99	R	-	FS	123	1	FSB	27	1	00	FA35	A869	10:19:01
23545	1428863	250	93	R	-	FS	123	1	FSB	27	1	00	381C	0EAE	10:19:03
23545	1428811	250	85	R	-	FS	125	1	FSB	27	1	00	5B58	CF3D	10:19:05
23545	1428766	250	78	R	-	FS	125	1	FSB	27	1	08	4FA2	7072	10:19:07
23545	1428724	250	72	R	-	FS	125	1	FSB	27	1	00	EC6D	EC6D	10:19:09
23545	1428687	250	64	R	-	FS	125	1	FSB	27	1	00	851C	5288	10:19:11
23545	1428653	250	57	R	-	FS	125	1	FSB	27	1	00	775E	15E2	10:19:13
23545	1428618	250	50	R	-	FS	127	3	FSB	27	1	08	1632	4801	10:19:15
23545	1428597	250	43	R	-	FS	127	3	FSB	27	1	00	64F8	0C23	10:19:17
23545	1428574	250	37	R	-	FS	127	3	-	27	1	00	86EE	BAC4	10:19:19
23545	1428556	250	32	R	-	FS	127	3	-	27	1	00	5E4A	DA76	10:19:21
23545	1428538	250	32	R	-	FS	127	3	-	27	1	08	10A4	3853	10:19:23
23545	1428520	250	32	R	-	FS	127	3	-	27	1	00	A88F	5189	10:19:25
23545	1428502	250	32	R	-	FS	127	3	-	27	1	00	A93D	1437	10:19:27
23545	1428484	250	32	R	-	FS	127	3	-	27	1	00	A2DD	1819	10:19:29
23545	1428466	250	32	R	-	FS	127	3	-	27	1	08	6E92	F091	10:19:31
23545	1428449	250	31	R	-	FS	127	3	-	27	1	00	F308	15D1	10:19:33
23545	1428431	250	31	R	-	FS	127	3	-	27	1	00	4780	69D3	10:19:35
23545	1428414	250	31	R	-	FS	127	3	-	27	1	00	8CAE	8FD1	10:19:37
23545	1428396	250	30	R	-	FS	127	3	-	27	1	08	009D	1F5C	10:19:39
23545	1428379	250	30	R	-	FS	127	3	-	27	1	00	CA8F	8671	10:19:41
23545	1428362	250	30	R	-	FS	127	3	-	27	1	00	EC4F	7C41	10:19:43
23545	1428345	250	30	R	-	FS	127	3	-	27	1	00	707A	8729	10:19:45
23545	1428328	250	30	R	-	FS	127	3	-	27	1	08	5BE7	D65D	10:19:47
23545	1428311	250	30	R	-	FS	127	3	-	27	1	00	F7E3	1D0D	10:19:49
23545	1428294	250	30	R	-	FS	127	3	-	27	1	00	81C5	AD53	10:19:51
23545	1428277	250	30	R	-	FS	127	3	-	27	1	00	1845	43EF	10:19:53
23545	1428261	250	29	R	-	FS	127	3	-	27	1	08	B747	4589	10:19:55

Figure: FSB applied and speed controlled to follow the speed limit (30 kmph)

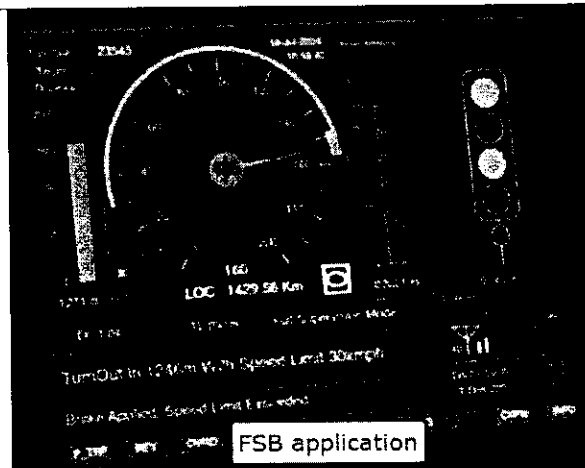


Figure: LP-OCIP showing application of FSB



Figure: LP-OCIP showing that speed limit in loop line was followed

#	Day1 - July 18 th Trial Details	
1	Date	18/07/24
2	Time	12:30 to 14:30 hrs
3	Loco ID	23546
4	Loco Configuration	Vande Bharat DTC 8 Coaches
5	Loco Type	Vande Bharat (HBL)
6	Division	NCR
7	Trial Section	VRBD - RDE (DN Direction)

1. Check SPAD prevention while communication failure from LTCAS(with LS Ack)

Test Scenario	Input Specification	Expected Result	Actual Result
Ensure Loco TCAS is registered with the Hodal Station TCAS and travelling in FS Mode at max section speed.	1) Railway staff performs the operation to keep Home signal S2 to Red aspect. 2) Remove the COM ports from the Radio modem-1 and Radio modem-2 in the LTCAS. 3) Loco pilot shall press ACK button	1) Loco TCAS shall maintain FS mode with display of speed, signal aspect and Movement Authority on LP-OCIP. 2) HBL personnel shall remove the radio com ports in the LTCAS. 3) Loco shall enter into LS Mode after 10 sec. (10 seconds timeout if radios disconnected and communication lost when loco is in the auto section) 4) FSB shall be applied after 7 sec of non acknowledgment.	1) This SPAD was conducted at Home signal S2. 2) FSB was applied by LTCAS (ABS LOC 1447.651) at a loco speed of 156 kmph when MA was 1413 mtrs to the home signal. 3) Simultaneously, radio comm ports were disconnected in the LTCAS when loco was about 250 mtrs from Down auto signal S144806. 4) After 10 sec Loco switched to LS Mode and "ACK LS Mode – Radio Comm Fail with Station" message was displayed on LP –

		5) Later when LS mode is acknowledged, FSB shall be released.	OCIP at ABS LOC 1448.04. 5) FSB was released as acknowledgment of LS mode was given by LP . 6) Radios were connected again when Loco was around 120 mtrs from the Home signal, and communication was established at AbsLoc 1448.86 7) Loco travelled about 1195 mtrs during the communication failure with the station and stopped at ABS LOC 1448.924 which is 53 mtrs before the Home signal.
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Locold	Abs Location	Train Len	Train Speed	Dir	Emergency	Mode	Tag-Id	TIN	BrakeStatus	StPrfid	StPrptoMA	HlthBits	RL	CRC	Time stamp
23546	1447315	250	157	N	-	FS	756	104	-	45	1	00	1728	71AB	13:05:37
23546	1447359	250	157	N	-	FS	756	104	-	45	1	00	1728	8F1F	13:05:39
23546	1447446	250	157	N	-	FS	756	104	-	45	1	00	1728	0244	13:05:41
23546	1447573	250	156	N	-	FS	758	104	-	45	1	00	1728	4633	13:05:43
23546	1447680	250	155	N	-	FS	758	104	FSB	45	1	00	1728	29FD	13:05:45
23546	1447703	250	156	N	-	FS	758	104	FSB	45	1	00	1728	0091	13:05:47
23546	1447792	250	154	N	-	FS	760	104	FSB	45	1	00	1728	F14A	13:05:49
23546	1447881	250	148	N	-	FS	760	104	FSB	45	1	00	1728	8527	13:05:51
23546	1447962	250	142	N	-	FS	760	104	FSB	45	1	00	1728	75DD	13:05:53
23546	1448034	250	134	N	-	LS	762	104	FSB	45	1	00	1C36	8A27	13:05:55
23546	1448107	250	127	N	-	LS	762	104	FSB	45	1	00	1C36	1DA6	13:05:57
23546	1448175	250	119	N	-	LS	762	104	FSB	45	1	00	1C36	95A9	13:05:59
23546	1448240	250	113	N	-	LS	762	104	FSB	45	1	00	1C36	0597	13:06:01
23546	1448329	250	102	N	-	LS	762	104	FSB	45	1	00	1C36	687C	13:06:03
23546	1448384	250	94	N	-	LS	762	104	FSB	45	1	00	1C36	3C8C	13:06:05
23546	1448435	250	74	N	-	LS	762	104	FSB	45	1	00	1C36	F982	13:06:07
23546	1448481	250	74	N	-	LS	762	104	FSB	45	1	00	1C36	F4A9	13:06:09
23546	1448524	250	66	N	-	LS	762	104	FSB	45	1	00	1C36	6D33	13:06:11
23546	1448563	250	59	N	-	LS	762	104	FSB	45	1	00	1C36	8DC1	13:06:13
23546	1448598	250	53	N	-	LS	762	104	FSB	45	1	00	1C36	9C81	13:06:15
23546	1448629	250	49	N	-	LS	762	104	FSB	45	1	00	1C36	1B75	13:06:17
23546	1448664	250	43	N	-	LS	762	104	FSB	45	1	00	1C36	8CF0	13:06:19
23546	1448699	250	32	N	-	LS	762	104	FSB	45	1	00	1C36	F920	13:06:21
23546	1448700	250	27	N	-	LS	764	104	-	45	1	00	1C36	5FCF	13:06:23
23546	1448732	250	26	N	-	LS	764	104	-	45	1	00	1C36	0275	13:06:25
23546	1448740	250	26	N	-	LS	764	104	-	45	1	00	1C36	0698	13:06:27
23546	1448755	250	26	N	-	LS	764	104	-	45	1	00	1C36	8889	13:06:29
23546	1448777	250	25	N	-	LS	764	104	-	45	1	00	1C36	6360	13:06:31
23546	1448790	250	23	N	-	LS	764	104	-	45	1	00	1C36	08C7	13:06:33
23546	1448797	250	22	N	-	LS	764	104	-	45	1	00	1C36	173E	13:06:35
23546	1448808	250	19	N	-	LS	764	104	-	45	1	00	1C36	5313	13:06:37
23546	1448819	250	17	N	-	LS	764	104	-	45	1	00	1C36	2AB2	13:06:39
23546	1448829	250	17	N	-	LS	764	104	-	45	1	00	1C36	5F1E	13:06:41

Abs Location	Train Len	Train Speed	Dir	Emergency	Mode	Tag-Id	TIN	BrakeStatus	StPrfid	StPrptoMA	HlthBits	RL	CRC	Time stamp
1448847	250	14	N	-	LS	764	104	-	45	1	00	1C36	9797	13:06:45
1448855	250	12	N	-	LS	764	104	-	45	1	00	1C36	6D89	13:06:47
1448862	250	12	N	-	FS	764	104	-	49	1	00	1C36	9881	13:06:49
1448871	250	11	N	-	FS	764	104	-	49	1	00	1C36	060A	13:06:51
1448877	250	10	N	-	FS	764	104	-	49	1	00	1C36	37EC	13:06:53
1448883	250	10	N	-	FS	764	104	-	49	1	00	1C36	57C6	13:06:55
1448889	250	10	N	-	FS	764	104	-	49	1	00	1C36	A7E1	13:06:57
1448895	250	10	N	-	FS	764	104	-	49	1	00	1C36	4145	13:06:59
1448901	250	10	N	-	FS	764	104	-	49	1	00	1C36	6896	13:07:01
1448906	250	10	N	-	FS	764	104	-	49	1	00	1C36	D986	13:07:03
1448911	250	6	N	-	FS	764	104	-	49	1	00	1C36	0753	13:07:05
1448913	250	6	N	-	FS	764	104	-	49	1	00	1C36	A5F8	13:07:07
1448916	250	5	N	-	FS	764	104	-	49	1	00	1C36	981F	13:07:09
1448921	250	4	N	-	FS	764	104	-	49	1	00	1C36	0D3C	13:07:11
1448923	250	3	N	-	FS	764	104	-	49	1	00	1C36	3C9A	13:07:13
1448923	250	2	N	-	FS	764	104	-	49	1	00	1C36	153A	13:07:15
1448924	250	0	N	-	FS	764	104	-	49	1	00	1C36	0FDE	13:07:17
1448924	250	0	N	-	FS	764	104	-	49	1	00	1C36	17FE	13:07:19
1448924	250	0	N	-	FS	764	104	-	49	1	00	1C36	5F1E	13:07:21

Figure: FSB applied to prevent SPAD, then loco switched to LS mode due to radio comm failure and finally loco stopped to prevent SPAD.

Text	TimeStamp
Ack LS Mode - Radio Comm Fail with Station	13:05:55
Vande Bharat DTC 8 Coaches	18/07/2024 13:05
LS Mode Waiting for Radio Comm with Station	13:06:01

Figure: DMIText Log showing the "ACK LS Mode – Radio Comm Fail with Station" message

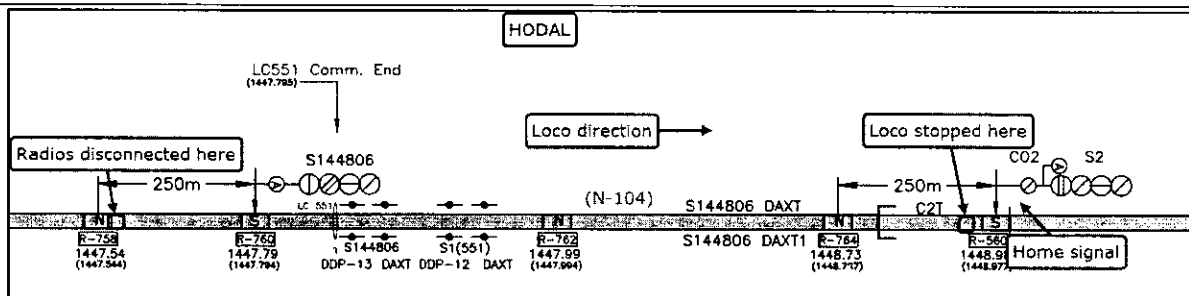


Figure: HODAL layout

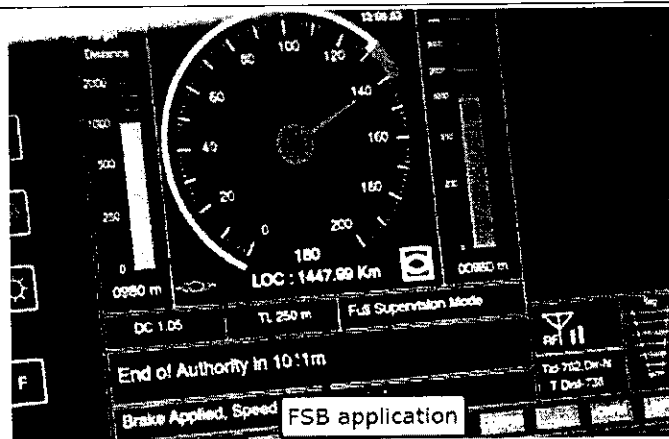


Figure: LP-OCIP showing FSB applied to prevent SPAD



Figure: "ACK LS Mode – Radio Comm Fail with Station" message was displayed on LP – OCIP



Figure:LP-OCIP showing Loco halted 53 mtrs before the Signal S2.

2. Verification of loop line speed control

Test Scenario	Input Specification	Expected Result	Actual Result
Loco TCAS is registered with the Station TCAS and travelling in FS Mode and approaching towards Sholaka loop line	Loco is travelling at maximum section speed on approach of Sholaka (SHLK) loop line in FS mode	At Sholaka first loop line, the permissible speed is 30 kmph. Loco TCAS shall apply brakes to reduce the speed within the permissible speed (30 kmph) in the turnout when Loco is in FS mode, the same can be seen on LP-OCIP.	- Loco was travelling at a speed of 158 kmph in DN direction. - FSB was applied at a speed of 158 Kmph in UP direction @Abs Loc 1458.845 and the Loco speed controlled to 29 kmph on reaching the Turn Out point in FS mode @AbsLoc 1460.093 - After entering the loop line, the loco maintained the speed below 30kmph.

Abs Location	Train Len	Train Speed	Dir	Emergency	Mode	Tag-Id	TIN	BrakeStatus	StPrfId	StPrptoMA	HlthBits	MAC	CRC	Time stamp
1458669	250	158	N	-	FS	476	100	-	1	1	00	CD8F 09F2	13:12:37	
1458757	250	158	N	-	FS	476	100	-	1	1	00	D676 BAA3	13:12:39	
1458845	250	158	N	-	FS	476	100	FSB	1	1	00	2628 EF49	13:12:41	
1458933	250	158	N	-	FS	476	100	FSB	1	1	00	9348 2005	13:12:43	
1459022	250	155	N	-	FS	478	100	FSB	1	1	00	3382 79F8	13:12:45	
1459106	250	148	N	-	FS	478	100	FSB	1	1	00	C645 66B6	13:12:47	
1459182	250	140	N	-	FS	480	100	FSB	1	1	00	0119 84E8	13:12:49	
1459259	250	133	N	-	FS	282	100	FSB	1	1	00	CC57 65B0	13:12:51	
1459329	250	127	N	-	FS	326	100	FSB	1	1	00	FA4F 1517	13:12:53	
1459398	250	119	N	-	FS	326	100	FSB	1	1	00	CBC2 D6CE	13:12:55	
1459464	250	113	N	-	FS	284	46	FSB	1	1	00	CCD1 DA26	13:12:57	
1459525	250	106	N	-	FS	284	46	FSB	1	1	00	E827 2532	13:12:59	
1459582	250	99	N	-	FS	284	46	FSB	1	1	00	98DE 6C2C	13:13:01	
1459635	250	93	N	-	FS	284	46	FSB	1	1	00	A0E4 9BED	13:13:03	
1459685	250	87	N	-	FS	284	46	FSB	1	1	00	F241 6397	13:13:05	
1459732	250	80	N	-	FS	284	46	FSB	1	1	00	F806 31EE	13:13:07	
1459781	250	74	N	-	FS	286	46	FSB	1	1	00	BF66 EE1A	13:13:09	
1459820	250	67	N	-	FS	286	46	FSB	1	1	00	F691 AD5F	13:13:11	
1459856	250	60	N	-	FS	286	46	FSB	1	1	00	C852 998E	13:13:13	
1459888	250	53	N	-	FS	286	46	FSB	1	1	00	F139 D76A	13:13:15	
1459915	250	46	N	-	FS	286	46	FSB	1	1	00	5A62 2180	13:13:17	
1459939	250	39	N	-	FS	286	46	-	1	1	00	71CE 9382	13:13:19	
1459962	250	32	N	-	FS	288	48	-	1	1	00	95C6 977C	13:13:21	
1459980	250	30	N	-	FS	288	48	-	1	1	00	D838 AAE9	13:13:23	
1459997	250	30	N	-	FS	288	48	-	1	1	00	9C46 B3F1	13:13:25	
1460014	250	29	N	-	FS	288	48	-	1	1	00	924A ADC8	13:13:27	
1460030	250	30	N	-	FS	288	48	-	1	1	00	957B ECE0	13:13:29	
1460047	250	30	N	-	FS	288	48	-	1	1	00	2876 21C4	13:13:31	
1460064	250	30	N	-	FS	288	48	-	1	1	00	F855 482C	13:13:33	
1460080	250	29	N	-	FS	288	48	-	1	1	00	9260 0A19	13:13:35	
1460097	250	29	N	-	FS	288	48	-	1	1	00	4140 0865	13:13:37	
1460113	250	28	N	-	FS	288	48	-	1	1	00	72B4 EED8	13:13:39	
1460129	250	28	N	-	FS	288	48	-	1	1	00	C43C 8A52	13:13:41	
1460145	250	29	N	-	FS	288	48	-	1	1	00	8BF3 BD0C	13:13:43	
1460161	250	28	N	-	FS	288	48	-	1	1	00	0A54 04F8	13:13:45	
1460177	250	28	N	-	FS	288	48	-	1	1	00	2045 0A8D	13:13:47	
1460193	250	28	N	-	FS	288	48	-	1	1	00	11E0 83A7	13:13:49	
1460209	250	28	N	-	FS	288	48	-	1	1	00	4E9F 33F4	13:13:51	

Figure: LocoRegPkts shows the application of FSB and speed reduced to 29kmph before loopline entry point

TimeStamp	LocoId	FrameOffset	Loco-Sos	TL Ref FNo	TL Ref Offset	TL-Type	Cur-Sig	Cur-Aspct	Nxt-Aspct	MA	AprStnId	AprSigDist	TO Spd
13:12:15	23546	0	0	-	-	0	Dn AS	YY	YY	3309	0	613	30
13:12:19	23546	0	0	-	-	0	Dn AS	YY	YY			437	30
13:12:27	23546	0	0	-	-	0	Dn AS	YY	YY	2780	0	84	30
13:12:27	23546	0	0	-	-	0	Dn H	YY	Y+1	2610	0	914	30
13:12:31	23546	0	0	-	-	0	Dn H	YY	Y+1	2434	0	738	30
13:12:35	23546	0	0	-	-	0	Dn H	YY	Y+1	2434	0	738	30
13:12:39	23546	0	0	-	-	0	Dn H	YY	Y+1	2257	0	561	30
13:12:43	23546	0	0	-	-	0	Dn H	YY	Y+1	2081	0	385	30
13:12:43	23546	0	0	-	-	0	Dn H	YY	Y+1	2081	0	385	30
13:12:43	23546	0	0	-	-	0	Dn H	YY	Y+1	1904	0	208	30
13:12:47	23546	0	0	-	-	0	Dn H	YY	Y+1	1904	0	208	30
13:12:51	23546	0	0	-	-	0	Dn H	YY	Y+1	1744	0	48	30
13:12:51	23546	0	0	-	-	0	Dn H	YY	Y+1	1744	0	48	30
13:12:55	23546	0	0	-	-	0	Dn rH ...	Y+1	R	1597	0	631	30
13:12:55	23546	0	0	-	-	0	Dn rH ...	Y+1	R	1597	0	631	30
13:12:59	23546	0	0	-	-	0	Dn rH ...	Y+1	R	1597	0	631	30

Figure: STCAS sent TurnOut speed limit as 30 which was followed by Loco

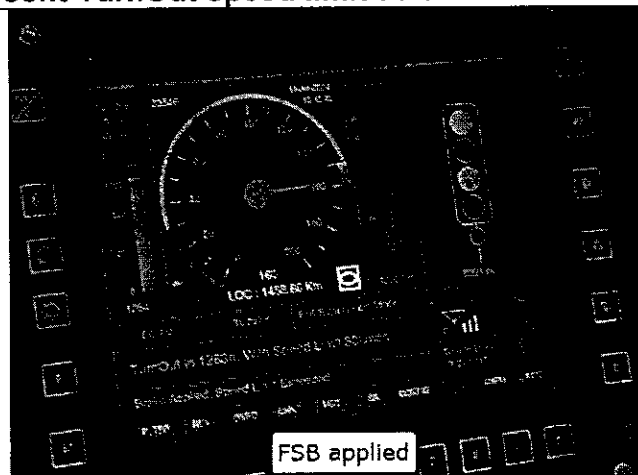


Figure: LP-OCIP shows that FSB was applied by Loco to attain speed limit

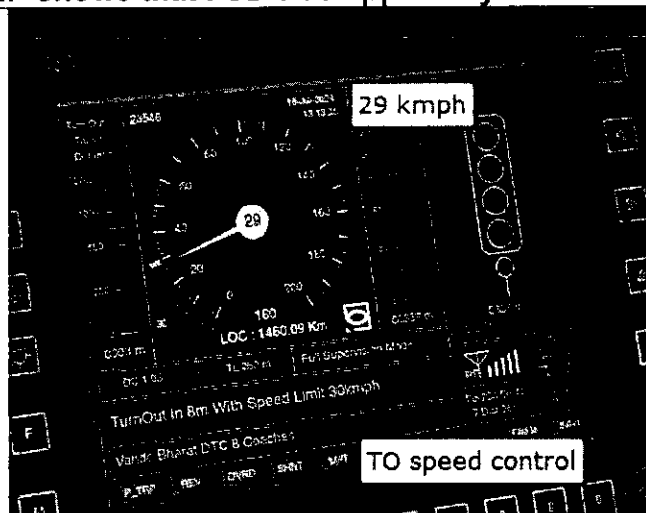


Figure: LP-OCIP shows that Loco entered the loopline with speed under 30 kmph

3. Check SPAD prevention while communication failure from LTCAS (without LS Ack)

Test Scenario	Input Specification	Expected Result	Actual Result
Ensure Loco TCAS is registered with the Rundhi Station TCAS and travelling in FS Mode at max section speed.	2) Railway staff performs the operation to keep Home signal S2 to Red aspect. 2) Remove the COM ports from the Radio modem-1 and Radio modem-2 in the LTCAS. 3) Do not press ACK button.	1) Loco TCAS shall maintain FS mode with display of speed, signal aspect and Movement Authority on LP-OCIP. 2) HBL personnel shall remove the radio com ports in the LTCAS. 3) Loco shall enter into LS Mode after 10 sec. (10 seconds timeout if radios disconnected and communication lost when loco is in the auto section) 4) FSB shall be applied after 7 sec of non-acknowledgment.	1) This SPAD was conducted at Home signal S2. 2) FSB was applied by LTCAS (ABS LOC 1468.259) at a loco speed of 158 kmph when MA was 1317 mtrs to the Home signal. 3) Simultaneously, radio comm ports were disconnected in the LTCAS when loco was about 550 mtrs from Down auto signal S146721. 4) After 10 sec Loco switched to LS Mode and "ACK LS Mode – Radio Comm Fail with Station" message was displayed on LP – OCIP at ABS LOC. 5) The previous FSB (for preventing SPAD by controlling speed before end of authority) continued to apply due to non-acknowledgment of LS mode by LP within 7 secs time window. 6) ACK for LS mode was given by LP at 7 kmph speed at

			AbsLoc 1469.42 when Loco was already decelerating.FSB was released due to ack given but still loco stopped 156 mtrs before the home signal. 7) Loco travelled about 1161 mtrs during the communication failure with the station and stopped at ABS LOC 1469.420 which is 156 mtrs before the Home signal.
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Day 1 – RDSO trials report (NCR)

HBL®

LocoId	Abs Location	Train Len	Train Speed	Dir	Emergency	Mode	Tag-Id	TIN	BrakeStatus	StPrfId	StPrpToMA	HlthBits	RL	CRC	Time stamp
23546	1468038	250	158	N	-	FS	190	96	-	9	1	00	1C36	85B7	13:21:43
23546	1468127	250	158	N	-	FS	190	96	-	9	1	00	1C36	8373	13:21:45
23546	1468171	250	158	N	-	FS	190	96	-	9	1	00	1C36	690B	13:21:47
23546	1468259	250	158	N	-	FS	190	96	FSB	9	1	00	1C36	27CF	13:21:49
23546	1468372	250	158	N	-	FS	192	96	FSB	9	1	00	1C36	26A4	13:21:51
23546	1468438	250	155	N	-	FS	192	96	FSB	9	1	00	1C36	7553	13:21:53
23546	1468522	250	148	N	-	FS	194	96	FSB	9	1	00	1C36	C3C7	13:21:55
23546	1468599	250	141	N	-	LS	194	96	FSB	9	1	00	1FF6	F4B9	13:21:57
23546	1468675	250	133	N	-	LS	194	96	FSB	9	1	00	1FF6	947D	13:21:59
23546	1468747	250	127	N	-	LS	194	96	FSB	9	1	00	1FF6	5EA8	13:22:01
23546	1468849	250	116	N	-	LS	194	96	FSB	9	1	00	1FF6	C96D	13:22:03
23546	1468911	250	109	N	-	LS	194	96	FSB	9	1	00	1FF6	E0AB	13:22:05
23546	1468976	250	102	N	-	LS	194	96	FSB	9	1	00	1FF6	06D2	13:22:07
23546	1469025	250	95	N	-	LS	194	96	FSB	9	1	00	1FF6	57AD	13:22:09
23546	1469076	250	90	N	-	LS	194	96	FSB	9	1	00	1FF6	D86A	13:22:11
23546	1469124	250	84	N	-	LS	194	96	FSB	9	1	00	1FF6	58A8	13:22:13
23546	1469173	250	76	N	-	LS	194	96	FSB	9	1	00	1FF6	8A91	13:22:15
23546	1469214	250	70	N	-	LS	194	96	FSB	9	1	00	1FF6	6EB5	13:22:17
23546	1469233	250	66	N	-	LS	194	96	FSB	9	1	00	1FF6	2200	13:22:19
23546	1469265	250	60	N	-	LS	194	96	FSB	9	1	00	1FF6	EADD	13:22:21
23546	1469311	250	51	N	-	LS	194	96	FSB	9	1	00	1FF6	43EA	13:22:23
23546	1469337	250	43	N	-	LS	194	96	FSB	9	1	00	1FF6	FA3	13:22:25
23546	1469351	250	39	N	-	LS	194	96	FSB	9	1	00	1FF6	01E	13:22:27
23546	1469383	250	32	N	-	LS	196	96	FSB	9	1	00	1FF6	49C1	13:22:29
23546	1469405	250	20	N	-	LS	196	96	FSB	9	1	00	1FF6	9DE1	13:22:31
23546	1469414	250	13	N	-	LS	196	96	FSB	9	1	00	1FF6	C41E	13:22:33
23546	1469417	250	9	N	-	LS	196	96	FSB	9	1	00	1FF6	88FE	13:22:35
23546	1469420	250	7	N	-	LS	196	96	FSB	9	1	00	1FF6	9D44	13:22:37
23546	1469420	250	0	N	-	LS	196	96	FSB	9	1	00	1FF6	2365	13:22:39
23546	1469420	250	0	N	-	LS	196	96	FSB	9	1	00	1FF6	9AC5	13:22:41
23546	1469420	250	0	N	-	LS	196	96	FSB	9	1	00	1FF6	4064	13:22:43
23546	1469420	250	0	N	-	LS	196	96	FSB	9	1	00	1FF6	AA85	13:22:45

Figure: FSB applied to prevent SPAD, then loco switched to LS mode due to radio comm failure, finally loco stopped before the signal and prevented SPAD.

Text	TimeStamp
End of Authority in 1066m	13:21:55
Ack LS Mode - Radio Comm Fail with Station	13:21:56
Vande Bharat DTC 8 Coaches	13:21:56
LS Mode Waiting for Radio Comm with Station	13:22:02
Brake Applied	Speed Limit Exceeded
Ack LS Mode - Radio Comm Fail with Station	13:22:08
LS Mode Waiting for Radio Comm with Station	13:22:13
Ack LS Mode - Radio Comm Fail with Station	13:22:19
LS Mode Waiting for Radio Comm with Station	13:22:25
Ack LS Mode - Radio Comm Fail with Station	13:22:31
LS Mode Waiting for Radio Comm with Station	13:22:37

Figure: DMIText Log showing the “ACK LS Mode – Radio Comm Fail with Station” message

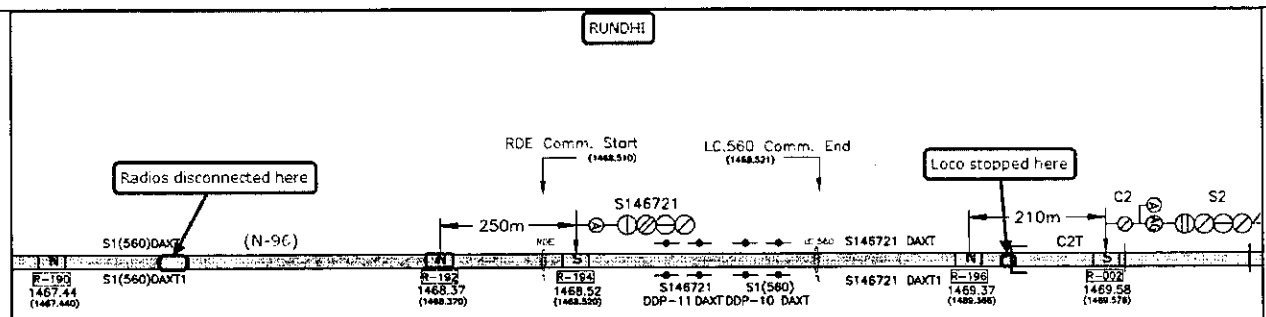


Figure: RUNDHI station layout



Figure: FSB applied to prevent SPAD



Figure: "ACK LS Mode – Radio Comm Fail with Station" message was displayed on LP – OCIP

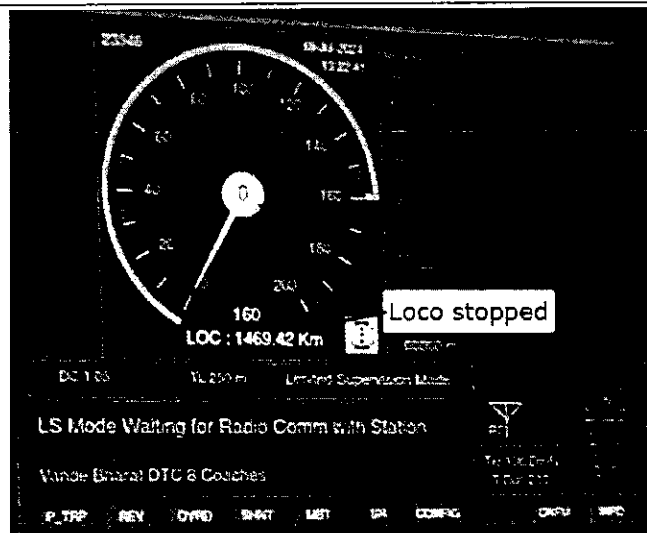


Figure: Loco halted 156 mtrs before the Signal S2.

#	Day2 - July 20 th Trial Details	
1	Date	20/07/24
2	Time	09:30 to 11:45 hrs
3	Loco ID	23545
4	Loco Configuration	Vande Bharat DTC 8 Coaches
5	Loco Type	Vande Bharat (HBL)
6	Division	NCR
7	Trial Section	RDE - VRBD (UP Direction)

1. Check SPAD prevention

Test Scenario	Input Specification	Expected Result	Actual Result
Ensure Loco TCAS is registered with the Sholak Station TCAS and travelling in FS Mode at max section speed.	Railway staff performs the operation to keep Home signal S75 to Red aspect.	Status of the Home signal S75 shall be displayed on LP-OCIP (and MA shall be displayed up to the Home signal)	1) This SPAD was conducted at Home signal S75 . 2) FSB was applied by LTCAS at a loco speed of 157 kmph when MA was 1326 mtrs to the Home signal at ABS LOC 1463.406
	Run the Loco with maximum section speed.	Loco TCAS shall apply brakes and Loco shall stop before the Home signal.	3) Loco applied FSB up to speed reduction 30 kmph. Then FSB was released at AbsLoc 1462.282 KM and then Loco moved at 15 kmph for a few meters. 4) Loco re-applied FSB (at AbsLoc 1462.107) and halted at ABS LOC 1462.085 which is 05 mtrs before the Home signal.

Locold	Abs Location	Train Len	Train Speed	Dir	Emergency	Mode	Tag-Id	TIN	BrakeStatus	StPrfId	StPrptoMA	HlthBits	RL	CRC	Time stamp
23545	1463468	250	157	R	-	FS	193	95	-	10	1	00	2C57	7F26	09:49:59
23545	1463380	250	158	R	-	FS	193	95	FSB	10	1	00	2C57	868C	09:50:01
23545	1463380	250	158	R	-	FS	193	95	FSB	10	1	00	2C57	868C	09:50:01
23545	1463208	250	156	R	-	FS	195	95	FSB	3	1	08	2C57	3C29	09:50:03
23545	1463208	250	156	R	-	FS	195	95	FSB	3	1	08	2C57	3C29	09:50:03
23545	1463145	250	151	R	-	FS	195	95	FSB	3	1	00	2C57	5185	09:50:05
23545	1463145	250	151	R	-	FS	195	95	FSB	3	1	00	2C57	5185	09:50:05
23545	1463145	250	151	R	-	FS	195	95	FSB	3	1	00	2C57	5185	09:50:05
23545	1463145	250	151	R	-	FS	195	95	FSB	3	1	00	2C57	5185	09:50:05
23545	1463064	250	144	R	-	FS	195	95	FSB	3	1	00	2C57	92A1	09:50:07
23545	1463064	250	144	R	-	FS	195	95	FSB	3	1	00	2C57	92A1	09:50:07
23545	1462986	250	137	R	-	FS	195	95	FSB	3	1	00	2C57	9D5B	09:50:09
23545	1462986	250	137	R	-	FS	195	95	FSB	3	1	00	2C57	9D5B	09:50:09
23545	1462912	250	129	R	-	FS	195	95	FSB	3	1	08	2C57	17D4	09:50:11
23545	1462912	250	129	R	-	FS	195	95	FSB	3	1	08	2C57	17D4	09:50:11
23545	1462842	250	122	R	-	FS	195	95	FSB	3	1	00	2C57	7063	09:50:13
23545	1462842	250	122	R	-	FS	195	95	FSB	3	1	00	2C57	7063	09:50:13
23545	1462776	250	115	R	-	FS	195	95	FSB	3	1	00	2C57	F6C2	09:50:15
23545	1462776	250	115	R	-	FS	195	95	FSB	3	1	00	2C57	F6C2	09:50:15
23545	1462714	250	108	R	-	FS	195	95	FSB	3	1	00	2C57	780D	09:50:17
23545	1462714	250	108	R	-	FS	195	95	FSB	3	1	00	2C57	780D	09:50:17
23545	1462656	250	100	R	-	FS	195	95	FSB	3	1	08	2C57	1986	09:50:19
23545	1462656	250	100	R	-	FS	195	95	FSB	3	1	08	2C57	1986	09:50:19
23545	1462602	250	94	R	-	FS	195	95	FSB	3	1	00	2C57	9520	09:50:21
23545	1462602	250	94	R	-	FS	195	95	FSB	3	1	00	2C57	9520	09:50:21
23545	1462576	250	92	R	-	FS	195	95	FSB	3	1	00	2C57	EB33	09:50:23
23545	1462576	250	92	R	-	FS	195	95	FSB	3	1	00	2C57	EB33	09:50:23
23545	1462527	250	85	R	-	FS	195	95	FSB	3	1	00	2C57	7345	09:50:25
23545	1462527	250	85	R	-	FS	195	95	FSB	3	1	00	2C57	7345	09:50:25
23545	1462461	250	75	R	-	FS	195	95	FSB	3	1	08	2C57	9FCA	09:50:27
23545	1462461	250	75	R	-	FS	195	95	FSB	3	1	08	2C57	9FCA	09:50:27
23545	1462421	250	68	R	-	FS	195	95	FSB	3	1	00	2C57	F839	09:50:29
23545	1462421	250	68	R	-	FS	195	95	FSB	3	1	00	2C57	F839	09:50:29
23545	1462402	250	65	R	-	FS	195	95	FSB	3	1	00	2C57	F116	09:50:31

Figure: FSB applied by Loco as per LocoRegPkts Log

Locold	Abs Location	Train Len	Train Speed	Dir	Emergency	Mode	Tag-Id	TIN	BrakeStatus	StPrfId	StPrptoMA	HlthBits	RL	CRC	Time stamp
23545	1462117	250	15	R	-	FS	199	95	-	3	1	00	2C57	DC5F	09:51:17
23545	1462112	250	14	R	-	FS	199	95	-	3	1	00	2C57	B578	09:51:19
23545	1462112	250	14	R	-	FS	199	95	-	3	1	00	2C57	B578	09:51:19
23545	1462104	250	14	R	-	FS	199	95	FSB	3	1	00	2C57	899C	09:51:21
23545	1462104	250	14	R	-	FS	199	95	FSB	3	1	00	2C57	899C	09:51:21
23545	1462095	250	14	R	-	FS	199	95	FSB	3	1	08	2C57	AF39	09:51:23
23545	1462096	250	14	R	-	FS	199	95	FSB	3	1	08	2C57	AF39	09:51:23
23545	1462089	250	9	R	-	FS	199	95	FSB	3	1	00	2C57	32D7	09:51:25
23545	1462089	250	9	R	-	FS	199	95	FSB	3	1	00	2C57	32D7	09:51:25
23545	1462086	250	3	R	-	FS	199	95	FSB	3	1	00	2C57	3EF4	09:51:27
23545	1462086	250	3	R	-	FS	199	95	FSB	3	1	00	2C57	3EF4	09:51:27
23545	1462085	250	0	R	-	FS	199	95	-	3	1	00	2C57	6C49	09:51:29
23545	1462085	250	0	R	-	FS	199	95	-	3	1	00	2C57	6C49	09:51:29
23545	1462085	250	0	R	-	FS	199	95	-	3	1	08	2C57	B6E8	09:51:31
23545	1462085	250	0	R	-	FS	199	95	-	3	1	08	2C57	B6E8	09:51:31
23545	1462085	250	0	R	-	FS	199	95	-	3	1	00	2C57	5C09	09:51:33
23545	1462085	250	0	R	-	FS	199	95	-	3	1	00	2C57	5C09	09:51:33
23545	1462085	250	0	R	-	FS	199	95	-	3	1	00	2C57	4429	09:51:35
23545	1462085	250	0	R	-	FS	199	95	-	3	1	00	2C57	4429	09:51:35
23545	1462085	250	0	R	-	FS	199	95	-	3	1	00	2C57	0CC9	09:51:37

Figure: FSB re-applied and Loco stopped and prevented SPAD

Abs Ltn	Gen SOS	LocoCount	Crc	TimeStamp	Locold	FrameOffset	Loco-Sos	TL Ref FNo	TL Ref Offset	TL-Type	Cur-Sig	Cur-Aspct	Nxt-Aspct	MA	AprStnId	AprSigDis
1460720	0	1	A6	09:51:31												
1460720	0	1	1A72	09:51:33	23545	1	0			0	Up H	R		5	0	5
1471144	0	0	9522	09:51:33	23545	1	0			0	Up H	R		5	0	5
1460720	0	1	1181	09:51:35												
1460720	0	1	1181	09:51:35	23545	1	0			0	Up H	R		5	0	5
1460720	0	1	A415	09:51:37	23545	1	0			0	Up H	R		5	0	5
1460720	0	1	32C8	09:51:39	23545	1	0			0	Up H	R		5	0	5
1460720	0	1	32C8	09:51:39	23545	1	0			0	Up H	R		5	0	5
1460720	0	1	517C	09:51:41	23545	1	0			0	Up H	R		5	0	5
1471144	0	0	5D	09:51:41												
1460720	0	1	A808	09:51:43	23545	1	0			0	Up H	R		5	0	5
1460720	0	1	A808	09:51:43	23545	1	0			0	Up H	R		5	0	5
1460720	0	1	3491	09:51:45	23545	1	0			0	Up H	R		5	0	5
1471144	0	0	9D	09:51:45												
1460720	0	1	771C	09:51:47	23545	1	0			0	Up H	R		5	0	5
1460720	0	1	771C	09:51:47	23545	1	0			0	Up H	R		5	0	5
1460720	0	1	D838	09:51:49	23545	1	0			0	Up H	R		5	0	5
1460720	0	1	7B09	09:51:51	23545	1	0			0	Up H	R		5	0	5
1460720	0	1	7B09	09:51:51	23545	1	0			0	Up H	R		5	0	5

Figure: MA and current signal after Loco stopped as per StnRegPkts Log.

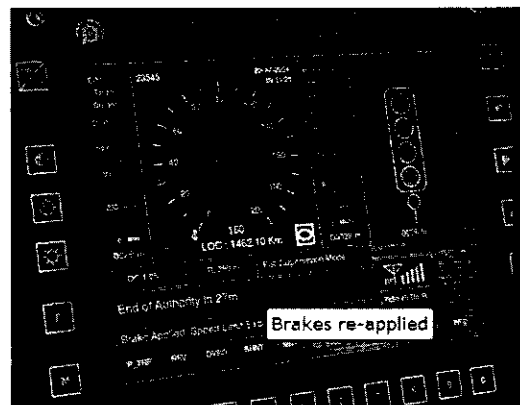
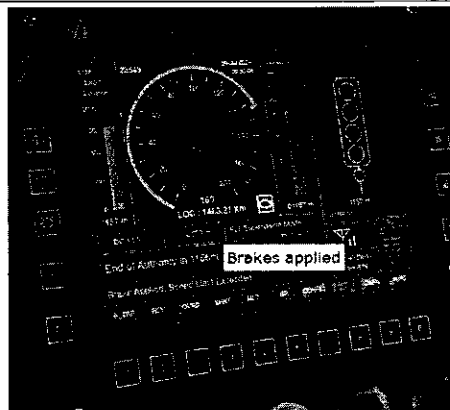


Figure: Brakes applied at various times to prevent SPAD as shown in LP-OCIP

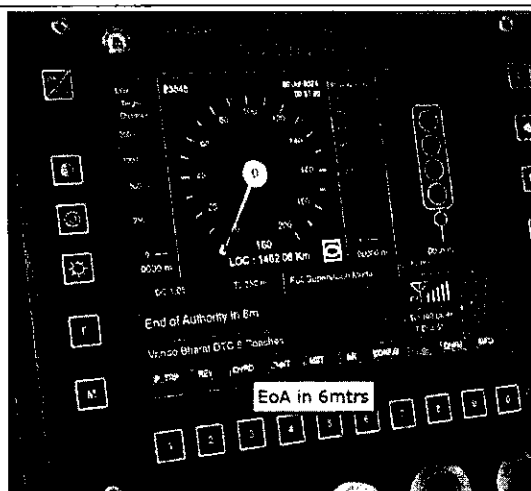


Figure: LP-OCIP showing that SPAD was prevented as Loco stopped before signal

2. Verification of loop line speed control

Test Scenario	Input Specification	Expected Result	Actual Result
Loco TCAS is registered with the Station TCAS and travelling in FS Mode and approaching towards Hodal (HDL) loop line	Loco is travelling at maximum section speed on approach of Hodal (HDL) loop line in FS mode	At Hodal first loop line, the permissible speed is 30 kmph. Loco TCAS shall apply brakes to reduce the speed within the permissible speed (30 kmph) in the turnout when Loco is in FS mode, the same can be seen on LP-OCIP.	- Loco was travelling at a speed of 160 kmph in UP direction. - FSB applied at a speed of 158 Kmph in UP direction @Abs Loc 1452.268 and the Loco speed controlled to 30 kmph on reaching the Turn Out point in FS mode @AbsLoc 1450.989 - After entering the loop line, the loco maintained the speed below 30kmph.

Locold	Abs Location	Train Len	Train Speed	Dir	Emergency	Mode	Tag-Id	TIN	BrakeStatus	StPrfid	StPrptoMA	HthBits	MAC	CRC	Time stamp
23545	1452356	250	158	R	-	FS	479	99	FSB	52	1	00	82C4	4455	09:56:57
23545	1452268	250	158	R	-	FS	479	99	FSB	52	1	08	8057	5F45	09:56:59
23545	1452170	250	157	R	-	FS	481	99	FSB	52	1	00	0533	98FF	09:57:01
23545	1452083	250	153	R	-	FS	481	99	FSB	52	1	00	3823	0E80	09:57:03
23545	1452000	250	146	R	-	FS	481	99	FSB	52	1	00	B208	4318	09:57:05
23545	1451908	250	140	R	-	FS	611	99	FSB	52	1	00	4A03	B4C5	09:57:07
23545	1451853	250	133	R	-	FS	611	99	FSB	52	1	00	F235	95AE	09:57:09
23545	1451774	250	126	R	-	FS	563	1	FSB	52	1	00	EB79	D383	09:57:11
23545	1451706	250	120	R	-	FS	563	1	FSB	52	1	00	0432	8E42	09:57:13
23545	1451641	250	113	R	-	FS	563	1	FSB	52	1	08	6C2B	9873	09:57:15
23545	1451579	250	107	R	-	FS	563	1	FSB	52	1	00	DAC8	A28A	09:57:17
23545	1451521	250	101	R	-	FS	563	1	FSB	52	1	00	F9E0	86A9	09:57:19
23545	1451467	250	94	R	-	FS	563	1	FSB	52	1	00	083E	8264	09:57:21
23545	1451416	250	88	R	-	FS	563	1	FSB	52	1	08	6367	9E08	09:57:23
23545	1451368	250	82	R	-	FS	565	1	FSB	52	1	00	D394	548F	09:57:25
23545	1451323	250	76	R	-	FS	565	1	FSB	52	1	00	17FF	87A8	09:57:27
23545	1451283	250	69	R	-	FS	567	3	FSB	52	1	00	C8AB	367B	09:57:29
23545	1451246	250	62	R	-	FS	567	3	FSB	52	1	08	09C0	8800	09:57:31
23545	1451214	250	55	R	-	FS	567	3	FSB	52	1	00	A7FB	0ACA	09:57:33
23545	1451185	250	49	R	-	FS	567	3	FSB	52	1	00	F785	DA62	09:57:35
23545	1451160	250	41	R	-	FS	567	3	FSB	52	1	00	8DCD	DA76	09:57:37
23545	1451139	250	34	R	-	FS	567	3	-	52	1	08	41B7	5413	09:57:39
23545	1451116	250	30	R	-	FS	569	5	-	52	1	00	8531	15CC	09:57:41
23545	1451099	250	30	R	-	FS	569	5	-	52	1	00	548D	32A8	09:57:43
23545	1451081	250	30	R	-	FS	569	5	-	52	1	00	C17A	C06C	09:57:45
23545	1451064	250	30	R	-	FS	569	5	-	52	1	08	E161	53A1	09:57:47
23545	1451047	250	30	R	-	FS	569	5	-	52	1	00	CD22	9A16	09:57:49
23545	1451030	250	30	R	-	FS	569	5	-	52	1	00	363F	59AA	09:57:51
23545	1451013	250	30	R	-	FS	569	5	-	52	1	00	2377	400C	09:57:53
23545	1450996	250	30	R	-	FS	569	5	-	52	1	08	0E15	1A73	09:57:55
23545	1450980	250	30	R	-	FS	569	5	-	52	1	00	9A78	DD42	09:57:57
23545	1450963	250	30	R	-	FS	569	5	-	52	1	00	8011	5583	09:57:59
23545	1450946	250	29	R	-	FS	569	5	-	52	1	00	FCDA	CD38	09:58:01
23545	1450930	250	29	R	-	FS	569	5	-	52	1	08	9E6A	2B85	09:58:03
23545	1450913	250	29	R	-	FS	569	5	-	52	1	00	8C2D	5534	09:58:05
23545	1450897	250	29	R	-	FS	569	5	-	52	1	00	80C5	8B18	09:58:07
23545	1450886	250	29	R	-	FS	591	15	-	52	1	00	AC69	8AB5	09:58:09
23545	1450869	250	28	R	-	FS	591	15	-	52	1	08	3F08	2C9B	09:58:11
23545	1450853	250	28	R	-	FS	591	15	-	52	1	00	ECC7	1B2C	09:58:13
23545	1450837	250	29	R	-	FS	591	15	-	52	1	00	984F	1E34	09:58:15
23545	1450821	250	28	R	-	FS	591	15	-	52	1	00	6354	26DD	09:58:17
23545	1450805	250	28	R	-	FS	591	15	-	52	1	08	2AD0	E77F	09:58:19

Figure: FSB applied and loco entered LoopLine with 30kmph and speed remained under 30kmph

Day2 – ISA trials report (NCR)

HBL®

TimeStamp	LocoId	FrameOffset	Loco-Sos	TL Ref FNo	TL Ref Offset	TL-Type	Cur-Sig	Cur-Aspct	Nxt-Aspct	MA	AprStnid	AprSigDist	TO Spd	DiffDistTO
09:57:01	23545	0	0			0	Up H	YY	Y+1	2107	0	307	30	1279
09:57:01	23545	0	0			0	Up H	YY	Y+1	2009	0	209	30	1181
09:57:03	23545	0	0			0	Up H	YY	Y+1	1922	0	122	30	1094
09:57:03	23545	0	0			0	Up H	YY	Y+1	1922	0	122	30	1094
09:57:05	23545	0	0			0	Up rH	Y+1	R	1745	0	785	30	919
09:57:07	23545	0	0			0	Up rH	Y+1	R	1745	0	785	30	919
09:57:09	23545	1	0			0	Up rH	Y+1	R	1745	0	785	30	919
09:57:09	23545	1	0			0	Up rH	Y+1	R	1745	0	785	30	919
09:57:13	23545	0	0			0	Up rH	Y+1	R	1543	0	583	30	717
09:57:13	23545	0	0			0	Up rH	Y+1	R	1543	0	583	30	717
09:57:15	23545	0	0			0	Up rH	Y+1	R	1478	0	518	30	652
09:57:15	23545	0	0			0	Up rH	Y+1	R	1478	0	518	30	652
09:57:17	23545	0	0			0	Up rH	Y+1	R	1416	0	456	30	590
09:57:17	23545	0	0			0	Up rH	Y+1	R	1416	0	456	30	590
09:57:19	23545	0	0			0	Up rH	Y+1	R	1358	0	398	30	532
09:57:19	23545	0	0			0	Up rH	Y+1	R	1358	0	398	30	532
09:57:21	23545	0	0			0	Up rH	Y+1	R	1358	0	398	30	532
09:57:21	23545	0	0			0	Up rH	Y+1	R	1358	0	398	30	532

Figure: TurnOut Speed Limit sent as 30 kmph (as per StnRegPkts Log)

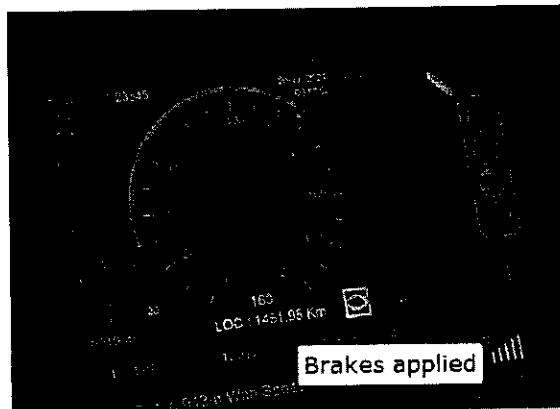


Figure: Brakes applied to control speed as shown in LP-OCIP

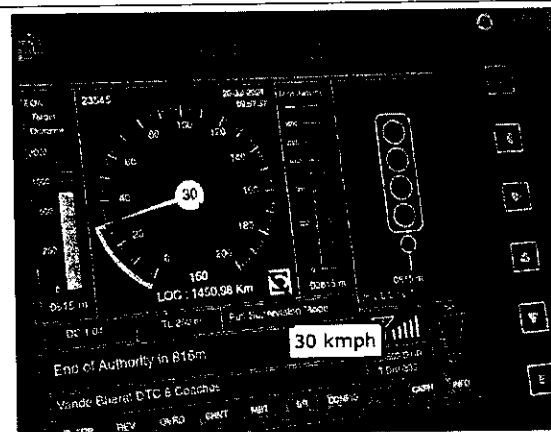


Figure: Loco entered Loop Line with speed 30 kmph as per LP-OCIP

3. Check FS mode to SR mode transition (without SR-Mode Ack) when tags are placed in wrong sequence

Test Scenario	Input Specification	Expected Result	Actual Result
Ensure Loco TCAS is healthy and registered with the Kosikalan station TCAS	Railway staff performs the operation at site to route to Main Line. HBL personnel shall swap any two tags on the main line (for which the route is set).	1) HBL personnel informed that the two tags #007 and #005 on the main line to be swapped (Both Main and duplicate tags). 2) After reading Tag – #005, loco shall switch to SR mode. 3) FSB shall be applied by the LTCAS after 7 secs of non-acknowledgment of SR MODE by LP.	1. Loco is travelling at max section speed in UP direction towards KSV station. 2. After crossing #03, loco read Tag-07 first and then Tag-05. 3. After reading #05, direction became un-defined (U) and loco switched to SR Mode and "ACK SR MODE – No section speed info" message was displayed on LP – OCIP. 4. LP did not acknowledge SR mode within the time window of 7 secs, FSB was applied by the LTCAS. 5. After reading the next Tag 09 loco switched to LS Mode and "Ack LS Mode" message was displayed on DMI which was not acknowledged. 6. As LS mode was not acknowledged, FSB continued to apply and Loco stopped at AbsLoc 1435.036.

7. Then, "Ack Block Section stop" message was shown on DMI and it was acknowledged within 4 seconds after which LP moved the Loco further without FSB application.

8. After reading second N-Tag #15, Loco switched to FS Mode and continued to travel.



VB 23545 Loco
log UP dir(200720

Loco Log attached for reference

Locold	Abs Location	Train Len	Train Speed	Dir	Emergency	Mode	Tag-Id	TIN	BrakeStatus	StPrfId	StPrptMA	HhthBits	RL	CRC	Time stamp
23545	1436195	250	110	R	-	FS	3	107	-	40	1	00	2F65	78BD	10:22:31
23545	1436195	250	110	R	-	FS	3	107	-	40	1	00	2F65	78BD	10:22:31
23545	1436127	250	111	R	-	FS	3	107	-	40	1	00	2F65	FEC1	10:22:33
23545	1436127	250	111	R	-	FS	3	107	-	40	1	00	2F65	FEC1	10:22:33
23545	1435838	250	111	R	-	FS	7	107	-	40	1	08	2F65	3CF4	10:22:35
23545	1435838	250	111	R	-	FS	7	107	-	40	1	08	2F65	3CF4	10:22:35
23545	1435776	250	111	R	-	FS	7	107	-	40	1	00	2F65	9E0A	10:22:37
23545	1435776	250	111	R	-	FS	7	107	-	40	1	00	2F65	9E0A	10:22:37
23545	1435714	250	111	R	-	FS	7	107	-	40	1	00	2F65	F256	10:22:39
23545	1435714	250	111	R	-	FS	7	107	-	40	1	00	2F65	F256	10:22:39
23545	1436050	250	111	U	-	SR	5	0	-	0	0	00	35F7	DA4E	10:22:41
23545	1436050	250	111	U	-	SR	5	0	-	0	0	00	35F7	DA4E	10:22:41
23545	0	250	109	U	-	SR	5	0	-	0	0	08	35F7	65B0	10:22:43
23545	0	250	109	U	-	SR	5	0	-	0	0	08	35F7	65B0	10:22:43
23545	0	250	109	U	-	SR	5	0	-	0	0	00	35F7	8F51	10:22:45
23545	0	250	109	U	-	SR	5	0	-	0	0	00	35F7	8F51	10:22:45
23545	0	250	109	U	-	SR	5	0	-	0	0	00	35F7	9771	10:22:47
23545	0	250	109	U	-	SR	5	0	-	0	0	00	35F7	9771	10:22:47
23545	1435600	250	109	R	-	SR	9	107	FSB	0	0	00	35F7	1E5C	10:22:49
23545	1435600	250	109	R	-	SR	9	107	FSB	0	0	00	35F7	1E5C	10:22:49
23545	1435560	250	109	R	-	SR	9	107	FSB	0	0	08	35F7	D0C0	10:22:51
23545	1435560	250	109	R	-	SR	9	107	FSB	0	0	08	35F7	D0C0	10:22:51
23545	1435493	250	105	R	-	SR	9	107	FSB	0	0	00	35F7	E851	10:22:53
23545	1435493	250	105	R	-	SR	9	107	FSB	0	0	00	35F7	E851	10:22:53
23545	1435464	250	103	R	-	SR	9	107	FSB	0	0	00	35F7	0C64	10:22:55
23545	1435464	250	103	R	-	SR	9	107	FSB	0	0	00	35F7	0C64	10:22:55
23545	1435409	250	95	R	-	SR	9	107	FSB	0	0	00	35F7	A2FF	10:22:57
23545	1435409	250	95	R	-	SR	9	107	FSB	0	0	00	35F7	A2FF	10:22:57
23545	1435409	250	95	R	-	SR	9	107	FSB	0	0	00	35F7	A2FF	10:22:57
23545	1435339	250	86	R	-	LS	9	107	FSB	34	1	08	35F7	4BA2	10:22:59
23545	1435339	250	86	R	-	LS	9	107	FSB	34	1	08	35F7	4BA2	10:22:59
23545	1435289	250	78	R	-	LS	9	107	FSB	34	1	00	35F7	8566	10:23:01
23545	1435289	250	78	R	-	LS	9	107	FSB	34	1	00	35F7	8566	10:23:01
23545	1435267	250	75	R	-	LS	9	107	FSB	34	1	00	35F7	F24B	10:23:03
23545	1435267	250	75	R	-	LS	9	107	FSB	34	1	00	35F7	F24B	10:23:03
23545	1435228	250	68	R	-	LS	9	107	FSB	34	1	00	35F7	49DD	10:23:05
23545	1435228	250	68	R	-	LS	9	107	FSB	34	1	00	35F7	49DD	10:23:05
23545	1435192	250	61	R	-	LS	9	107	FSB	34	1	08	35F7	0046	10:23:07
23545	1435192	250	61	R	-	LS	9	107	FSB	34	1	08	35F7	0046	10:23:07
23545	1435160	250	55	R	-	LS	9	107	FSB	34	1	00	35F7	64B1	10:23:09
23545	1435160	250	55	R	-	LS	9	107	FSB	34	1	00	35F7	64B1	10:23:09
23545	1435134	250	48	R	-	LS	9	107	FSB	34	1	00	35F7	8800	10:23:11
23545	1435134	250	48	R	-	LS	9	107	FSB	34	1	00	35F7	8800	10:23:11

Figure: Loco logs showing the SR mode and direction Undefined after tags were read in wrong sequence and then FSB applied, then Loco switched to LS mode.

Locold	Abs Location	Train Len	Train Speed	Dir	Emergency	Mode	Tag-Id	TIN	BrakeStatus	StPrfId	StPrptMA	HhthBits	RL	CRC	Time stamp
23545	1435106	250	43	R	-	LS	9	107	FSB	34	1	00	35F7	812F	10:23:13
23545	1435106	250	43	R	-	LS	9	107	FSB	34	1	00	35F7	812F	10:23:13
23545	1435085	250	36	R	-	LS	9	107	FSB	34	1	08	35F7	8098	10:23:15
23545	1435085	250	36	R	-	LS	9	107	FSB	34	1	08	35F7	8098	10:23:15
23545	1435067	250	28	R	-	LS	9	107	FSB	34	1	00	35F7	F646	10:23:17
23545	1435067	250	28	R	-	LS	9	107	FSB	34	1	00	35F7	F646	10:23:17
23545	1435054	250	22	R	-	LS	9	107	FSB	34	1	00	35F7	6A68	10:23:19
23545	1435054	250	22	R	-	LS	9	107	FSB	34	1	00	35F7	6A68	10:23:19
23545	1435043	250	14	R	-	LS	9	107	FSB	34	1	00	35F7	748C	10:23:21
23545	1435037	250	3	R	-	LS	9	107	FSB	34	1	08	35F7	0130	10:23:23
23545	1435037	250	3	R	-	LS	9	107	FSB	34	1	08	35F7	0130	10:23:23
23545	1435036	250	0	R	-	LS	9	107	FSB	34	1	00	35F7	D4CE	10:23:25
23545	1435036	250	0	R	-	LS	9	107	FSB	34	1	00	35F7	D4CE	10:23:25
23545	1435036	250	0	R	-	LS	9	107	FSB	34	1	00	35F7	CCEFF	10:23:27
23545	1435036	250	0	R	-	LS	9	107	FSB	34	1	00	35F7	CCEFF	10:23:27
23545	1435036	250	0	R	-	LS	9	107	FSB	34	1	00	35F7	754E	10:23:29
23545	1435036	250	0	R	-	LS	9	107	FSB	34	1	00	35F7	754E	10:23:29
23545	1435036	250	0	R	-	LS	9	107	FSB	34	1	08	35F7	AFFB	10:23:31
23545	1435036	250	0	R	-	LS	9	107	FSB	34	1	08	35F7	AFFB	10:23:31
23545	1435036	250	0	R	-	LS	9	107	FSB	34	1	00	35F7	450E	10:23:33
23545	1435036	250	0	R	-	LS	9	107	FSB	34	1	00	35F7	450E	10:23:33
23545	1435036	250	0	R	-	LS	9	107	FSB	34	1	00	35F7	5D2E	10:23:35
23545	1435036	250	0	R	-	LS	9	107	FSB	34	1	00	35F7	5D2E	10:23:35
23545	1435036	250	0	R	-	LS	9	107	FSB	34	1	00	35F7	258E	10:23:41
23545	1435036	250	0	R	-	LS	9	107	FSB	34	1	00	35F7	258E	10:23:41
23545	1435036	250	0	R	-	LS	9	107	FSB	34	1	00	35F7	3DAE	10:23:43
23545	1435036	250	0	R	-	LS	9	107	FSB	34	1	00	35F7	3DAE	10:23:43
23545	1435036	250	0	R	-	LS	9	107	FSB	34	1	00	35F7	E58F	10:23:45
23545	1435036	250	0	R	-	LS	9	107	FSB	34	1	00	35F7	E58F	10:23:45

Figure: FSB applied due to non acknowledgment of LS mode and Loco stopped (from Locoreq packet log)

Day2 - ISA trials report (NCR)

HBL

Locold	Abs Location	Train Len	Train Speed	Dir	Emergency	Mode	Tag-Id	TIN	BrakeStat	StPrfid	StPrptoMA	HhthBits	MAC	CRC	Time stamp
23545	1435036	250	0	R	-	LS	9	107	-	34	1	00	B528	40DB	10:23:27
23545	1435036	250	0	R	-	LS	9	107	-	34	1	00	1EEA	5885	10:23:29
23545	1435036	250	0	R	-	LS	9	107	-	34	1	08	741F	165CA	10:23:31
23545	1435036	250	0	R	-	LS	9	107	-	34	1	00	085E	8PCB	10:23:33
23545	1435036	250	0	R	-	LS	9	107	-	34	1	00	0801	C3FA	10:23:35
23545	1435036	250	0	R	-	LS	9	107	-	34	1	00	0852	9E3C	10:23:37
23545	1435036	250	0	R	-	LS	9	107	-	34	1	00	0862	6C62	10:23:39
23545	1435036	250	0	R	-	LS	9	107	-	34	1	00	5E52	8A74	10:23:41
23545	1435036	250	0	R	-	LS	9	107	-	34	1	00	D3AF	73F0	10:23:43
23545	1435036	250	0	R	-	LS	9	107	-	34	1	00	3D65	5C1C	10:23:45
23545	1435036	250	0	R	-	LS	9	107	-	34	1	08	0531	B9AF	10:23:47
23545	1435036	250	0	R	-	LS	9	107	-	34	1	00	1D9D	2E44	10:23:49
23545	1435036	250	0	R	-	LS	9	107	-	34	1	00	BAFB	2C9E	10:23:51
23545	1435036	250	0	R	-	LS	9	107	-	34	1	00	5C5C	BE75	10:23:53
23545	1435036	250	0	R	-	LS	9	107	-	34	1	08	86FF	7DCF	10:23:55
23545	1435035	250	2	R	-	LS	9	107	-	34	1	00	7035	90EC	10:23:57
23545	1435033	250	4	R	-	LS	9	107	-	34	1	00	EA69	6FA4	10:23:59
23545	1435029	250	7	R	-	LS	9	107	-	34	1	00	BEFB	39BA	10:24:01
23545	1435024	250	10	R	-	LS	9	107	-	34	1	08	7A8B	A24C	10:24:03
23545	1435017	250	13	R	-	LS	9	107	-	34	1	00	0F38	1AC5	10:24:05
23545	1435009	250	16	R	-	LS	9	107	-	34	1	00	702D	D606	10:24:07
23545	1434999	250	19	R	-	LS	9	107	-	34	1	00	D307	F07A	10:24:09
23545	1434987	250	22	R	-	LS	9	107	-	34	1	08	E855	B441	10:24:11
23545	1434974	250	22	R	-	LS	9	107	-	34	1	00	F77B	7830	10:24:13
23545	1434948	250	23	R	-	LS	11	107	-	34	1	00	7B5F	0979	10:24:15
23545	1434937	250	23	R	-	LS	11	107	-	34	1	00	1092	E280	10:24:17

Figure: After LS mode ack, Loco started to move forward without FSB application
(from locoregular packet log)

Locold	Abs Location	Train Len	Train Speed	Dir	Emergency	Mode	Tag-Id	TIN	BrakeStatus	StPrfid	StPrptoMA	HhthBits	RL	CRC	Time stamp
23545	1434161	250	86	R	-	LS	13	107	-	34	1	00	35F7	CF75	10:25:27
23545	1434112	250	89	R	-	LS	13	107	-	34	1	00	35F7	2489	10:25:29
23545	1434112	250	89	R	-	LS	13	107	-	34	1	00	35F7	2489	10:25:29
23545	1434062	250	92	R	-	LS	13	107	-	34	1	08	35F7	60A9	10:25:31
23545	1434062	250	92	R	-	LS	13	107	-	34	1	08	35F7	60A9	10:25:31
23545	1434010	250	94	R	-	LS	13	107	-	34	1	00	35F7	1C6F	10:25:33
23545	1434010	250	94	R	-	LS	13	107	-	34	1	00	35F7	1C6F	10:25:33
23545	1433964	250	96	R	-	FS	15	107	-	34	1	00	35F7	E3CA	10:25:35
23545	1433964	250	96	R	-	FS	15	107	-	34	1	00	35F7	E3CA	10:25:35
23545	1433909	250	99	R	-	FS	15	107	-	34	1	00	35F7	414E	10:25:37
23545	1433909	250	99	R	-	FS	15	107	-	34	1	00	35F7	414E	10:25:37
23545	1433847	250	102	R	-	FS	15	107	-	34	1	08	35F7	D038	10:25:39
23545	1433847	250	102	R	-	FS	15	107	-	34	1	08	35F7	D038	10:25:39
23545	1433790	250	104	R	-	FS	15	107	-	34	1	00	35F7	7EF4	10:25:41
23545	1433790	250	104	R	-	FS	15	107	-	34	1	00	35F7	7EF4	10:25:41
23545	1433728	250	107	R	-	FS	17	107	-	40	1	00	35F7	48CB	10:25:43
23545	1433728	250	107	R	-	FS	17	107	-	40	1	00	35F7	48CB	10:25:43
23545	1433728	250	107	R	-	FS	17	107	-	40	1	00	35F7	48CB	10:25:43
23545	1433728	250	107	R	-	FS	17	107	-	40	1	00	35F7	48CB	10:25:43
23545	1433676	250	109	R	-	FS	17	107	-	43	1	00	35F7	0A56	10:25:45
23545	1433578	250	110	R	-	FS	17	107	-	43	1	08	35F7	4111	10:25:47
23545	1433578	250	110	R	-	FS	17	107	-	43	1	08	35F7	4111	10:25:47

Figure: Loco logs showing that Loco switched to FS mode after second normal tag #15 was read

Text	TimeStamp	
Manned LC Gate 547 approaching in 328m	10:22:38 AM	
Manned LC Gate 547 approaching in 297m	10:22:39 AM	
Manned LC Gate 547 approaching in 266m	10:22:40 AM	Ack SR mode alert
Manned LC Gate 547 approaching in 235m	10:22:41 AM	Ack LS Mode alert
Ack SR Mode - No Section Speed Info	10:22:42 AM	
Brake Applied	Speed Limit Exceeded	10:22:48 AM
Ack LS Mode	10:22:58 AM	
Vande Bharat DTC 8 Coaches	10:23:25 AM	
Ack Block Section stop	SOS Generates in 6 s	10:23:29 AM
Ack Block Section stop	SOS Generates in 5 s	10:23:30 AM
Ack Block Section stop	SOS Generates in 4 s	10:23:31 AM
Manned LC Gate 546 approaching in 608m	10:25:26 AM	
Manned LC Gate 546 approaching in 584m	10:25:27 AM	Ack given here
Manned LC Gate 546 approaching in 559m	10:25:28 AM	
Manned LC Gate 546 approaching in 534m	10:25:29 AM	

Figure: DMI_text Log showing the Alert

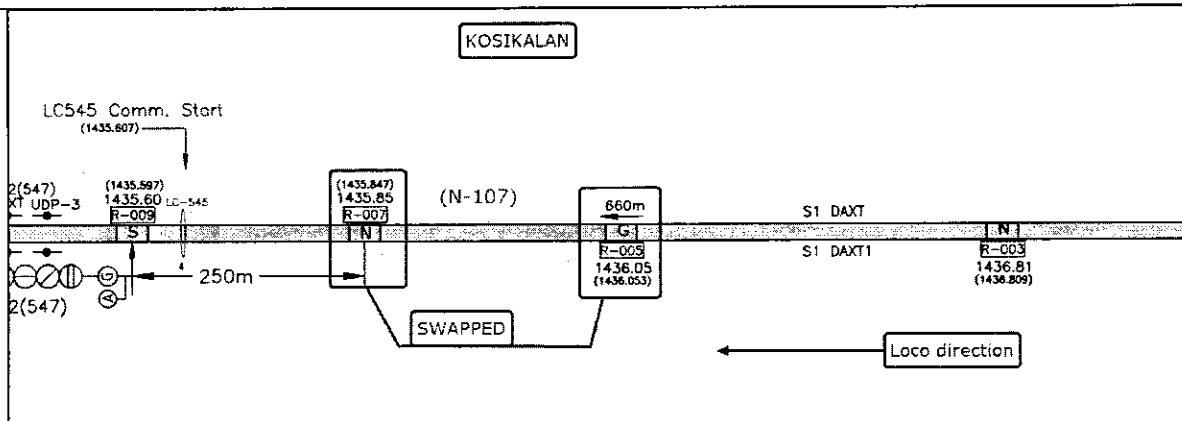


Figure: Kosikalan Layout showing the position of swapped tags in the field.

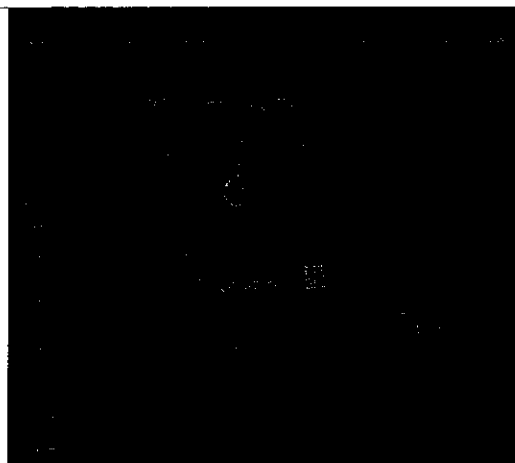


Figure: LP-OCIP showing that Loco switched to SR mode after reading tag#5 and direction became undefined, ack SR mode alert was shown on screen.

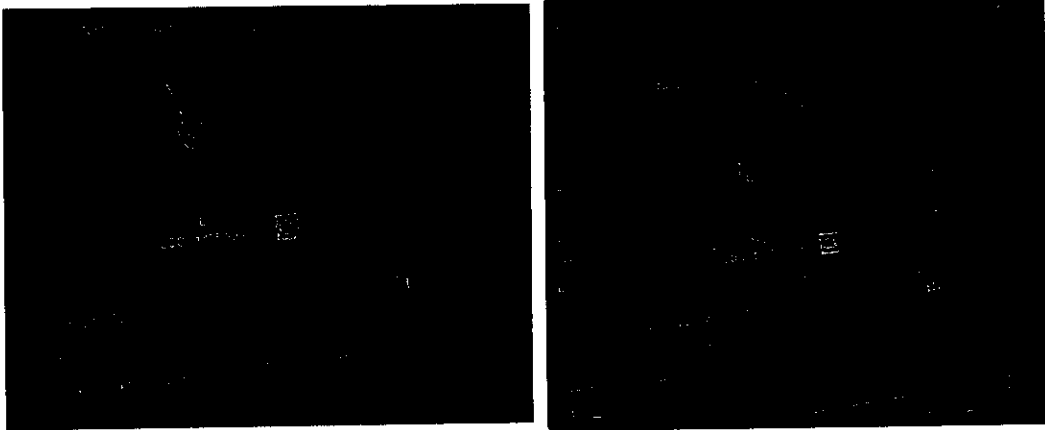


Figure: LP-OCIP showing that Loco switched to LS mode after reading tag#9 and also Loco got the direction. In the second figure, Loco switched to FS mode after reading second N-tag#15 (two normal tags #11 & 15 were read, #13 is a Gate tag).

4. Verification of loop line speed control

Test Scenario	Input Specification	Expected Result	Actual Result
Loco TCAS is registered with the Station TCAS and travelling in FS Mode and approaching towards Chhata loop line	Loco is travelling at maximum section speed on approach of Chhata (SHJ) loop line in FS mode	At Chhata first loop line, the permissible speed is 30 kmph. Loco TCAS shall apply brakes to reduce the speed within the permissible speed (30 kmph) in the turnout when Loco is in FS mode, the same can be seen on LP-OCIP.	- Loco was travelling at a speed of 137 kmph in UP direction. - FSB applied at a speed of 137 Kmph in UP direction @Abs Loc 1429.303 and the Loco speed controlled to 31 kmph on reaching the Turn Out point in FS mode @AbsLoc 1428.320 - After entering the loop line, the loco maintained the speed below 30kmph.

LocoId	Abs Location	Train Len	Train Speed	Dir	Emergency	Mode	Tag-Id	TIN	BrakeStatus	StPrId	StPrtoMA	HlthBits	MAC	CRC	Time stamp
23545	1429376	250	135	R	-	FS	121	107	-	27	1	00	8A81	69E0	10:28:23
23545	1429303	250	137	R	-	FS	121	107	FSB	27	1	00	9F9A	2F15	10:28:25
23545	1429208	250	137	R	-	FS	123	1	FSB	27	1	08	B285	7638	10:28:27
23545	1429151	250	134	R	-	FS	123	1	FSB	27	1	00	B6E6	02B6	10:28:29
23545	1429077	250	128	R	-	FS	123	1	FSB	27	1	00	236D	0FC1	10:28:31
23545	1429008	250	121	R	-	FS	123	1	FSB	27	1	00	D5E0	B850	10:28:33
23545	1428943	250	115	R	-	FS	123	1	FSB	27	1	08	BEA0	AC57	10:28:35
23545	1428881	250	107	R	-	FS	123	1	FSB	27	1	00	6198	9589	10:28:37
23545	1428818	250	100	R	-	FS	125	1	FSB	27	1	00	E806	D871	10:28:39
23545	1428767	250	92	R	-	FS	125	1	FSB	27	1	00	9030	D015	10:28:41
23545	1428717	250	85	R	-	FS	125	1	FSB	27	1	00	2B34	E00D	10:28:43
23545	1428671	250	78	R	-	FS	125	1	FSB	27	1	00	5768	8CE9	10:28:45
23545	1428618	250	71	R	-	FS	127	3	FSB	27	1	00	B51A	E162	10:28:47
23545	1428590	250	64	R	-	FS	127	3	FSB	27	1	00	4927	F829	10:28:49
23545	1428556	250	57	R	-	FS	127	3	FSB	27	1	08	0BAA	7868	10:28:51
23545	1428526	250	50	R	-	FS	127	3	FSB	27	1	00	9526	9EDC	10:28:53
23545	1428	Speed at LoopLine entry	32	R	-	FS	127	3	FSB	27	1	00	97E2	0EE3	10:28:55
23545	1428	33	R	-	FS	127	3	-	-	27	1	00	5AFC	17F1	10:28:57
23545	1428459	250	33	R	-	FS	127	3	-	27	1	08	4AC3	2B11	10:28:59
23545	1428440	250	32	R	-	FS	127	3	-	27	1	00	39A2	DC08	10:29:01
23545	1428422	250	32	R	-	FS	127	3	-	27	1	00	03AF	D9F1	10:29:03
23545	1428404	250	32	R	-	FS	127	3	-	27	1	00	14DA	2A28	10:29:05
23545	1428385	250	32	R	-	FS	127	3	-	27	1	08	B8E5	DAB9	10:29:07
23545	1428367	250	32	R	-	FS	127	3	OverSpd	27	1	00	C24D	DE65	10:29:09
23545	1428349	250	32	R	-	FS	127	3	OverSpd	27	1	00	3087	0DE2	10:29:11
23545	1428331	250	32	R	-	FS	127	3	OverSpd	27	1	00	29DD	AABA	10:29:13
23545	1428313	250	32	R	-	FS	127	3	OverSpd	27	1	08	48DC	8CD2	10:29:15
23545	1428296	250	31	R	-	FS	127	3	OverSpd	27	1	00	386E	2093	10:29:17
23545	1428278	250	31	R	-	FS	127	3	OverSpd	27	1	00	F1D0	3050	10:29:19
23545	1428260	250	30	R	-	FS	127	3	-	27	1	00	67D0	7399	10:29:21
23545	1428244	250	28	R	-	FS	127	3	-	27	1	08	BD33	6E3D	10:29:23
23545	1428228	250	28	R	-	FS	127	3	-	27	1	00	D933	FCFD	10:29:25
23545	1428208	250	28	R	-	FS	127	3	-	27	1	00	A842	C7EE	10:29:27
23545	1428192	250	28	R	-	FS	127	3	-	27	1	00	9286	0416	10:29:29
23545	1428176	250	28	R	-	FS	147	15	-	27	1	08	3448	6253	10:29:31
23545	1428160	250	28	R	-	FS	147	15	-	27	1	00	BCBD	771F	10:29:33
23545	1428145	250	27	R	-	FS	147	15	-	27	1	00	2111	5963	10:29:35
23545	1428129	250	28	R	-	FS	147	15	-	27	1	00	6EAF	20FC	10:29:37
23545	1428114	250	27	R	-	FS	147	15	-	27	1	08	B183	780C	10:29:39
23545	1428099	250	27	R	-	FS	147	15	-	27	1	00	0EE7	9882	10:29:41

Figure: LocoLog showing the application of FSB and speed at Loopline entry point

TimeStamp	Locid	FrameOffset	Loco-Sos	TL Ref FNo	TL Ref Offset	TL-Type	Cur-Sig	Cur-Aspct	Nxt-Aspct	MA	AprStnd	AprSigDist	TO Spd	DiffDistTO
10:28:23	23545	0	0	-	-	0	Up rH ...	Y+1	R	1866	0	760	30	1056
10:28:25														
10:28:25														
10:28:25														
10:28:25	23545	0	0	-	-	0	Up rH ...	Y+1	R	1793	0	687	30	983
10:28:25	23545	0	0	-	-	0	Up rH ...	Y+1	R	1793	0	687	30	983
10:28:27														
10:28:27														
10:28:27	23545	0	0	-	-	0	Up rH ...	Y+1	R	1698	0	592	30	888
10:28:29														
10:28:29														
10:28:29	23545	0	0	-	-	0	Up rH ...	Y+1	R	1641	0	535	30	831
10:28:29	23545	0	0	-	-	0	Up rH ...	Y+1	R	1641	0	535	30	831
10:28:31														
10:28:31														
10:28:31	23545	0	0	-	-	0	Up rH ...	Y+1	R	1567	0	461	30	757
10:28:31	23545	0	0	-	-	0	Up rH ...	Y+1	R	1567	0	461	30	757
10:28:33	23545	0	0	-	-	0	Up rH ...	Y+1	R	1498	0	392	30	688

Figure: TurnOut speed Limit 30 kmph sent by STCAS



Figure: Brakes applied to control speed as per LP-OCIP view



Figure: Speed controlled while entering Loopline as per LP-OCIP view

5. SPAD prevention while radio communication failure in STCAS (without LS Ack)

Test Scenario	Input Specification	Expected Result	Actual Result
Ensure Loco TCAS is registered with the Ajhai Station TCAS and travelling in FS mode.	1) Railway staff performs the operation to keep the Home signal S75 aspect as Red 2) Remove the COM ports from the Radio modem-1 and Radio modem-2 in the Station 3) Do not press ACK button	1) Loco TCAS shall maintain FS mode with display of speed, signal aspect and Movement Authority on LP- OCIP 2) HBL personnel shall remove the radio com ports in the STCAS. 3) Loco shall enter into <u>LS Mode</u> after 30 sec (30 seconds timeout if radios disconnected and communication lost when loco is in the station/block section) 4) FSB shall be applied after 7 sec	1) Loco was travelling at a speed of 160 kmph in UP direction. 2) SPAD prevention test performed at S75 signal. 3) FSB was applied at the Abs Loc 1420.586 Km @159 kmph with an MA of 1406 mtrs. 4) Simultaneously, radio comm ports were disconnected in the STCAS when Loco was about 924 mtrs from the Home Signal S75. 5) After 30 sec Loco switched to LS Mode and "ACK LS Mode – Radio Comm Fail with Station" message was displayed on LP – OCIP at Abs Loc 1419.487 Km. 6) FSB was again applied at AbsLoc 1419. 458 due to non-acknowledgment of LS mode by LP within 7 secs time window. 7) Loco halted at Abs Loc 1419.435 Km, which is 167 mtrs before the S75.

8) Loco travelled about 685 mtrs during the communication failure with the station until the loco stopped.

LocoId	Abs Location	Train Len	Train Speed	Dir	Emergency	Mode	Tag-Id	TIN	BrakeStatus	StPrfid	StPrptoMA	HlthBits	RL	CRC	Time stamp
23545	1420843	250	159	R	-	FS	329	111	-	22	1	00	3163	58EC	10:42:07
23545	1420754	250	159	R	-	FS	329	111	-	22	1	00	3163	58D2	10:42:09
23545	1420754	250	159	R	-	FS	329	111	-	22	1	00	3163	58D2	10:42:09
23545	1420621	250	159	R	-	FS	329	111	FSB	22	1	08	3163	5032	10:42:11
23545	1420621	250	159	R	-	FS	329	111	FSB	22	1	08	3163	5032	10:42:11
23545	1420533	250	158	R	-	FS	329	111	FSB	22	1	00	3163	3037	10:42:13
23545	1420533	250	158	R	-	FS	329	111	FSB	22	1	00	3163	3037	10:42:13
23545	1420498	250	158	R	-	FS	331	111	FSB	22	1	00	3163	9207	10:42:15
23545	1420498	250	158	R	-	FS	331	111	FSB	22	1	00	3163	9207	10:42:15
23545	1420404	250	154	R	-	FS	331	111	FSB	22	1	00	3163	1AA2	10:42:17
23545	1420404	250	154	R	-	FS	331	111	FSB	22	1	00	3163	1AA2	10:42:17
23545	1420298	250	148	R	-	FS	333	111	FSB	22	1	08	3163	6934	10:42:19
23545	1420298	250	148	R	-	FS	333	111	FSB	22	1	08	3163	6934	10:42:19
23545	1420254	250	140	R	-	FS	339	111	FSB	22	1	00	3163	67F6	10:42:21
23545	1420254	250	140	R	-	FS	339	111	FSB	22	1	00	3163	67F6	10:42:21
23545	1420185	250	134	R	-	FS	339	111	FSB	22	1	00	3163	F7FB	10:42:23
23545	1420185	250	134	R	-	FS	339	111	FSB	22	1	00	3163	F7FB	10:42:23
23545	1420113	250	126	R	-	FS	339	111	FSB	22	1	00	3163	CBBA	10:42:25
23545	1420113	250	126	R	-	FS	339	111	FSB	22	1	00	3163	CBBA	10:42:25
23545	1420044	250	119	R	-	FS	339	111	FSB	22	1	08	3163	5C38	10:42:27
23545	1420044	250	119	R	-	FS	339	111	FSB	22	1	08	3163	5C38	10:42:27
23545	1419980	250	112	R	-	FS	339	111	FSB	22	1	00	3163	5C38	10:42:29
23545	1419980	250	112	R	-	FS	339	111	FSB	22	1	00	3163	5C38	10:42:29
23545	1419914	250	105	R	-	FS	339	111	FSB	22	1	00	3163	1006	10:42:31
23545	1419914	250	105	R	-	FS	339	111	FSB	22	1	00	3163	1006	10:42:31
23545	1419858	250	97	R	-	FS	339	111	FSB	22	1	00	3163	A495	10:42:33
23545	1419858	250	97	R	-	FS	339	111	FSB	22	1	00	3163	A495	10:42:33
23545	1419780	250	88	R	-	FS	339	111	FSB	22	1	08	3163	211F	10:42:35

Figure: Loco log showing the application of FSB to prevent SPAD and when radios were disconnected

LocoId	Abs Location	Train Len	Train Speed	Dir	Emergency	Mode	Tag-Id	TIN	BrakeStatus	StPrfid	StPrptoMA	HlthBits	RL	CRC	Time stamp
23545	1419529	250	39	R	-	FS	339	111	FSB	22	1	00	3163	F400	10:42:49
23545	1419511	250	33	R	-	FS	335	111	FSB	22	1	08	3163	826A	10:42:51
23545	1419511	250	33	R	-	FS	335	111	FSB	22	1	08	3163	826A	10:42:51
23545	1419493	250	24	R	-	FS	335	111	-	22	1	00	3163	8F93	10:42:53
23545	1419493	250	24	R	-	FS	335	111	-	22	1	00	3163	8F93	10:42:53
23545	1419487	250	20	R	-	LS	335	111	-	22	1	00	32A9	74C7	10:42:55
23545	1419487	250	20	R	-	LS	335	111	-	22	1	00	32A9	74C7	10:42:55
23545	1419477	250	14	R	-	LS	335	111	-	22	1	00	32A9	0AE7	10:42:57
23545	1419477	250	14	R	-	LS	335	111	-	22	1	00	32A9	0AE7	10:42:57
23545	1419466	250	14	R	-	LS	335	111	-	22	1	08	32A9	C24D	10:42:59
23545	1419466	250	14	R	-	LS	335	111	FSB	22	1	00	32A9	C24D	10:42:59
23545	1419458	250	14	R	-	LS	335	111	FSB	22	1	00	32A9	D486	10:43:01
23545	1419458	250	14	R	-	LS	335	111	FSB	22	1	00	32A9	D486	10:43:01
23545	1419454	250	13	R	-	LS	335	111	FSB	22	1	00	32A9	65A1	10:43:03
23545	1419454	250	13	R	-	LS	335	111	FSB	22	1	00	32A9	65A1	10:43:03
23545	1419447	250	14	R	-	LS	335	111	FSB	22	1	00	32A9	EE44	10:43:05
23545	1419447	250	14	R	-	LS	335	111	FSB	22	1	08	32A9	EE44	10:43:05
23545	1419440	250	10	R	-	LS	335	111	FSB	22	1	08	32A9	6EE0	10:43:07
23545	1419440	250	10	R	-	LS	335	111	FSB	22	1	08	32A9	6EE0	10:43:07
23545	1419436	250	3	R	-	LS	335	111	FSB	22	1	00	32A9	C30A	10:43:09
23545	1419436	250	3	R	-	LS	335	111	FSB	22	1	00	32A9	C30A	10:43:09
23545	1419435	250	0	R	-	LS	335	111	-	22	1	00	32A9	D830	10:43:11
23545	1419435	250	0	R	-	LS	335	111	-	22	1	00	32A9	D830	10:43:11
23545	1419435	250	0	R	-	LS	335	111	-	22	1	00	32A9	A292	10:43:13
23545	1419435	250	0	R	-	LS	335	111	-	22	1	00	32A9	A292	10:43:13
23545	1419435	250	0	R	-	LS	335	111	-	22	1	08	32A9	7833	10:43:15
23545	1419435	250	0	R	-	LS	335	111	-	22	1	08	32A9	7833	10:43:15
23545	1419435	250	0	R	-	LS	335	111	-	22	1	00	32A9	92D2	10:43:17

Figure: Loco switched to LS(30 sec after radios disconnected) due to no communication with STCAS and FSB was re-applied due to non-acknowledgment of LS mode.

Figure: DMI_Text log showing LS mode and ACK alerts



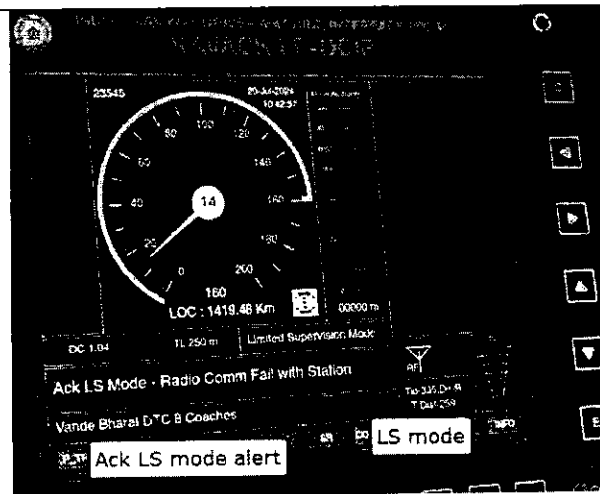


Figure: LS Mode and "ACK LS Mode " message was displayed

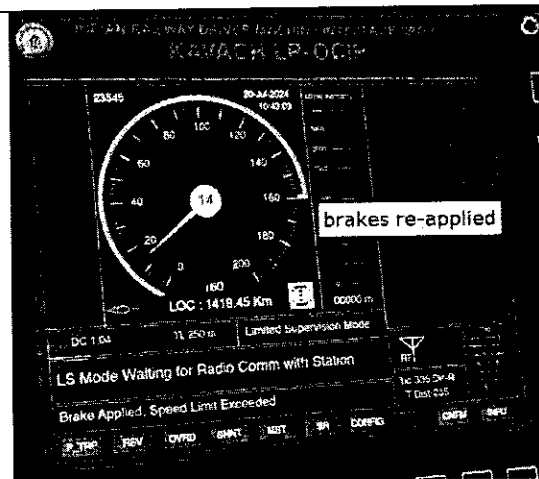


Figure: FSB re-applied due to non-acknowledgment of LS mode within 7 seconds.

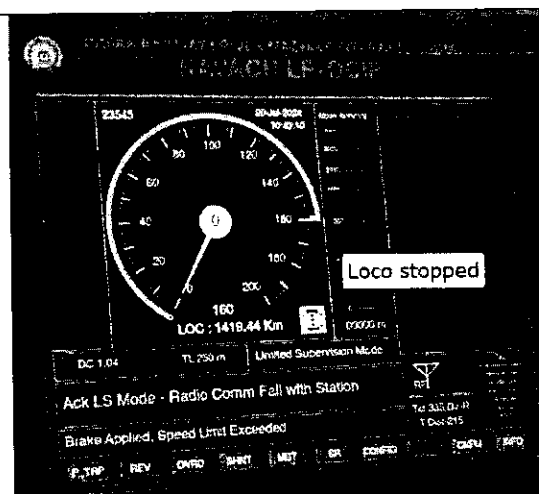


Figure: Loco stopped 167 mtrs before the signal and prevented SPAD.

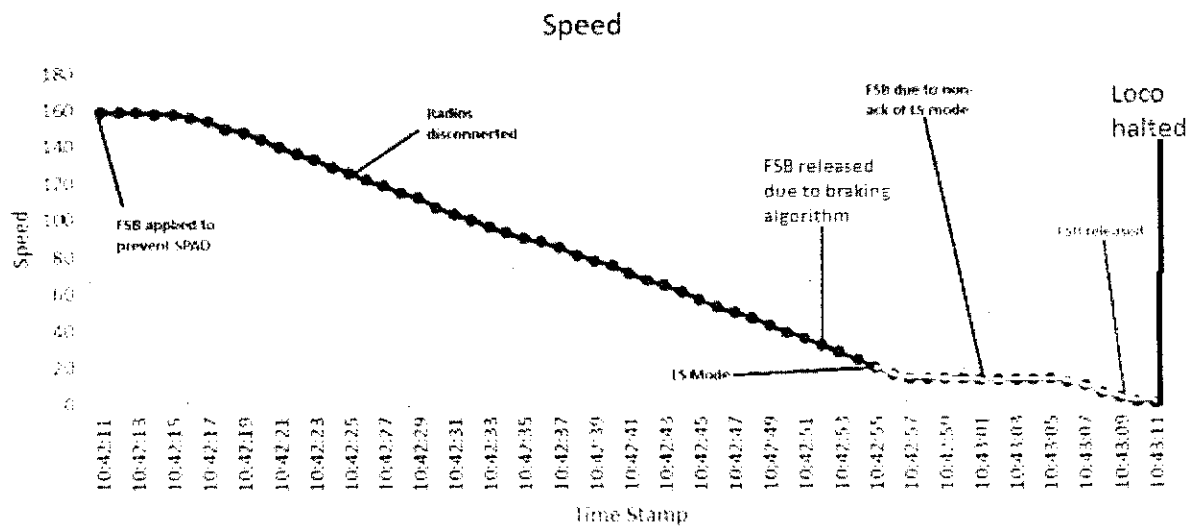


Figure: Speed vs Time stamp graph of Loco movement and braking



BRAKING
CURVE_AJhai.xlsx

Double click to open the above braking curve document (~sheet 1 and sheet 2)

6. Check SPAD prevention

Test Scenario	Input Specification	Expected Result	Actual Result
Ensure Loco TCAS is registered with the Vrindaban Station TCAS and travelling in FS Mode at max section speed.	Railway staff performs the operation to keep Home signal S75 to Red aspect.	Status of the Home signal S75 shall be displayed on LP-OCIP (and MA shall be displayed up to the home signal)	1) This SPAD was conducted at Home signal S75 . 2) FSB was applied by LTCAS at a loco speed of 115 kmph when MA was 768 mtrs to the Home signal at ABS LOC 1409.065
	Run the Loco with maximum section speed.	Loco TCAS shall apply brakes and Loco shall stop before the Home signal.	3) Loco applied FSB up to speed reduction 22 kmph. Then FSB was released at AbsLoc 1408.487 KM and the Loco moved at 14 kmph for a few meters. 4) Loco re-applied FSB (at AbsLoc 1408.360) and halted at ABS LOC 1408.346 which is 11 mtrs before the home signal.

TimeStamp	Locold	FrameOffset	Loco-Sos	TL Ref FNo	TL Ref Offset	TL-Type	Cur-Sig	Cur-Aspct	Nxt-Aspct	MA	AprStnId
10:59:09											
10:59:09	23545	1	0	-	-	0	Up H	R	-	11	0
10:59:11											
10:59:11	23545	1	0	-	-	0	Up H	R	-	11	0
10:59:11											
10:59:11	23545	1	0	-	-	0	Up H	R	-	11	0
10:59:13											
10:59:13	23545	1	0	-	-	0	Up H	R	-	11	0
10:59:13											
10:59:13	23545	1	0	-	-	0	Up H	R	-	11	0
10:59:15											
10:59:15	23545	1	0	-	-	0	Up H	R	-	11	0
10:59:15											
10:59:15	23545	1	0	-	-	0	Up H	R	-	11	0
10:59:17											
10:59:17	23545	1	0	-	-	0	Up H	R	-	11	0
10:59:17											
10:59:17	23545	1	0	-	-	0	Up H	R	-	11	0

Figure: StnRegPkts showing that SPAD was prevented by 11 mtrs.

Day2 - ISA trials report (NCR)

HBL

LocoId	Abs Location	Train Len	Train Speed	Dir	Emergency	Mode	Tag-Id	TIN	BrakeStatus	StPrfId	StPrptoMA	HlthBits	MAC	CRC	Time stamp
23545	1409166	250	111	R	-	FS	635	115	-	5	1	00	D41D	B680	10:57:57
23545	1409103	250	114	R	-	FS	635	115	FSB	5	1	00	A707	4853	10:57:59
23545	1409039	250	116	R	-	FS	635	115	FSB	5	1	00	9268	B6CE	10:58:01
23545	1408976	250	113	R	-	FS	635	115	FSB	5	1	00	283A	946A	10:58:03
23545	1408915	250	106	R	-	FS	635	115	FSB	5	1	00	8BE4	755C	10:58:05
23545	1408858	250	103	R	-	FS	635	115	FSB	5	1	00	CCFA	4486	10:58:07
23545	1408805	250	92	R	-	FS	635	115	FSB	5	1	00	78A3	5899	10:58:09
23545	1408756	250	85	R	-	FS	635	115	FSB	5	1	00	D945	C21C	10:58:11
23545	1408710	250	78	R	-	FS	635	115	FSB	5	1	00	39D2	67EF	10:58:13
23545	1408669	250	72	R	-	FS	635	115	FSB	5	1	00	36BD	93AC	10:58:15
23545	1408631	250	65	R	-	FS	635	115	FSB	5	1	00	ACDD	0048	10:58:17
23545	1408598	250	57	R	-	FS	635	115	FSB	5	1	00	9E98	C82A	10:58:19
23545	1408571	250	51	R	-	FS	637	115	FSB	5	1	00	72D7	B6F8	10:58:21
23545	1408544	250	44	R	-	FS	637	115	FSB	5	1	00	6D93	26DB	10:58:23
23545	1408522	250	37	R	-	FS	637	115	FSB	5	1	00	AA19	9A39	10:58:25
23545	1408503	250	30	R	-	FS	637	115	FSB	5	1	00	289D	396A	10:58:27
23545	1408488	250	24	R	-	FS	637	115	FSB	5	1	00	A917	6C61	10:58:29
23545	1408477	250	17	R	-	FS	637	115	-	5	1	00	54EB	4404	10:58:31
23545	1408469	250	14	R	-	FS	637	115	-	5	1	00	F8F1	BA1B	10:58:33
23545	1408461	250	14	R	-	FS	637	115	-	5	1	00	2799	0AEC	10:58:35
23545	1408453	250	14	R	-	FS	637	115	-	5	1	00	4806	7464	10:58:37
23545	1408445	250	14	R	-	FS	637	115	-	5	1	00	ADC8	8CBF	10:58:39
23545	1408437	250	14	R	-	FS	637	115	-	5	1	00	ESBE	2FAC	10:58:41
23545	1408430	250	14	R	-	FS	637	115	-	5	1	00	9A27	55E1	10:58:43
23545	1408422	250	14	R	-	FS	637	115	-	5	1	00	D0A6	43F3	10:58:45
23545	1408414	250	14	R	-	FS	637	115	-	5	1	00	428D	50E4	10:58:47
23545	1408407	250	14	R	-	FS	637	115	-	5	1	00	8C2D	9655	10:58:49
23545	1408399	250	14	R	-	FS	637	115	-	5	1	00	2EFD	F483	10:58:51
23545	1408391	250	13	R	-	FS	637	115	-	5	1	00	0A94	998D	10:58:53
23545	1408384	250	13	R	-	FS	637	115	-	5	1	00	8BE9	A639	10:58:55
23545	1408376	250	14	R	-	FS	637	115	-	5	1	00	A0BA	3222	10:58:57
23545	1408368	250	13	R	-	FS	637	115	-	5	1	00	D6A2	28A1	10:58:59
23545	1408361	250	13	R	-	FS	637	115	FSB	5	1	00	953D	B721	10:59:01
23545	1408354	250	12	R	-	FS	637	115	FSB	5	1	00	3DC5	5C79	10:59:03
23545	1408348	250	6	R	-	FS	637	115	FSB	5	1	00	763C	5073	10:59:05
23545	1408346	250	0	R	-	FS	637	115	-	5	1	00	A47B	8ED5	10:59:07
23545	1408346	250	0	R	-	FS	637	115	-	5	1	00	53D8	2D32	10:59:09
23545	1408346	250	0	R	-	FS	637	115	-	5	1	00	5E80	5F17	10:59:11
23545	1408346	250	0	R	-	FS	637	115	-	5	1	00	6C36	F57B	10:59:13
23545	1408346	250	0	R	-	FS	637	115	-	5	1	00	9EF0	DD42	10:59:15
23545	1408346	250	0	R	-	FS	637	115	-	5	1	00	EFD9	B946	10:59:17
23545	1408346	250	0	R	-	FS	637	115	-	5	1	00	D058	F1EA	10:59:19
23545	1408346	250	0	R	-	FS	637	115	-	5	1	00	8628	3D69	10:59:21

Figure: FSB applied at various times to prevent SPAD (as per Locoregular pkt log)

#	Day2 - July 20 th Trial Details	
1	Date	20/07/24
2	Time	12:30 to 14:30 hrs
3	Loco ID	23546
4	Loco Configuration	WAP-6
5	Loco Type	Vande Bharat DTC 8 Coaches
6	Division	NCR
7	Trial Section	VRBD - RDE (DN Direction)

1. SPAD prevention while radio communication failure from LTCAS (without LS Ack)

Test Scenario	Input Specification	Expected Result	Actual Result
Ensure Loco TCAS is registered with the Ajhai Station TCAS and travelling in FS Mode.	1) Railway staff performs the operation to keep the Home signal S2 aspect as Red 2) Remove the COM ports from the Radio modem-1 and Radio modem-2 in the LTCAS 3) Do not press ACK button	1) Loco TCAS shall maintain FS mode with display of speed, signal aspect and Movement Authority on LP- OCIP 2) HBL personnel shall remove the radio com ports in the LTCAS. 3) Loco shall enter into LS Mode after 10 sec(10 seconds timeout if radios disconnected and communication lost when loco is in the auto section) 4) FSB shall be applied after 7 sec of non-ack.	1) Loco was travelling at a speed of 160 kmph in Down direction. 2) SPAD prevention test performed at S2 signal. 3) FSB was applied at the Abs Loc 1414994 Km @158 Kmph with an MA of 1323 mtrs. 4) Radio comm ports were disconnected in the LTCAS when Loco was about 440 mtrs from the Down AutoSecGateStop signal S1(536). 5) After 10 sec Loco switched to LS Mode and "ACK LS Mode – Radio Comm Fail with Station" message was displayed on LP – OCIP at Abs Loc 1414.994 Km.

6) The FSB(for preventing SPAD by controlling speed before end of authority) applied due to non-acknowledgment of LS mode by LP within **7 secs** time window.

7) Loco halted at **Abs Loc 1416.049 Km**, which is **274 mtrs** before the **S2**.

8) Loco travelled about **1543 mtrs** during the communication failure with the station until the loco stopped.

TimeStamp	LocId	FrameOffset	Loco-Sos	TL Ref FNo	TL Ref Offset	TL-Type	Cur-Sig	Cur-Aspct	Nxt-Aspct	MA	AprStnId	AprSigDist	TO Spd	D/H Dist TO	TO Ref Dist
12:47:11	23546	1	0			0	Dn GS...	Y		1933	36012	553		0	0
12:47:13	23546	1	0			0	Dn GS...	Y		1817	36012	437		0	0
12:47:13	23546	1	0			0	Dn GS...	Y		1817	36012	437		0	0
12:48:19															
12:48:19															
12:48:21															
12:48:21															
12:48:21															
12:48:23															
12:48:25															
12:48:25															
12:48:27															
12:48:27	23546	1	0			0	Dn H	R		274	0	274		0	0
12:48:27	23546	1	0			0	Dn H	R		274	0	274		0	0

Figure: StnRegPkts Log showing no communication between LTCAS and STCAS when radios were disconnected.

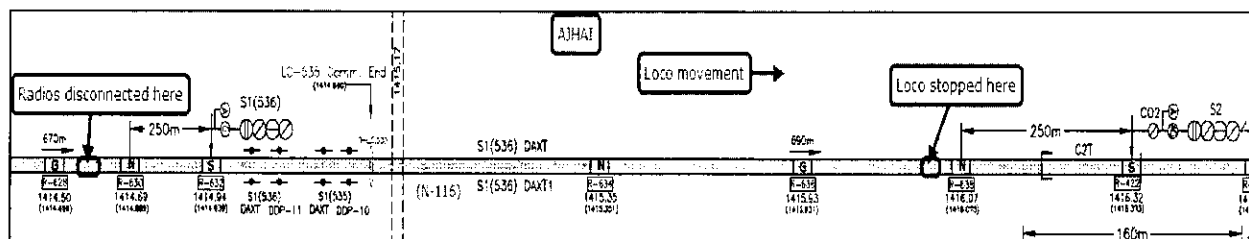


Figure: AJHAI station layout

Locold	Abs Location	Train Len	Train Speed	Dir	Emergency Mode	Tag-Id	TIN	BrakeStatus	StPrfId	StPrptoMA	HthBits	RL	CRC	Time stamp
23546	1414502	250	158	N	FS	628	116	-	11	1	00	097E	A9ED	12:47:13
23546	1414502	250	158	N	FS	628	116	-	11	1	00	097E	A9ED	12:47:13
23546	1414599	250	159	N	FS	628	116	-	11	1	08	097E	1A0D	12:47:15
23546	1414692	250	159	N	FS	630	116	-	11	1	02	097E	CEDA	12:47:17
23546	1414771	250	158	N	FS	630	116	-	11	1	00	097E	687F	12:47:19
23546	1414859	250	159	N	FS	630	116	-	11	1	00	097E	64D0	12:47:21
23546	1414924	250	158	N	LS	632	116	FSB	11	1	08	17DE	CC71	12:47:23
23546	1415082	250	156	N	LS	632	116	FSB	11	1	02	17DE	945D	12:47:25
23546	1415168	250	150	N	LS	632	116	FSB	11	1	00	17DE	2A82	12:47:27
23546	1415249	250	143	N	LS	632	116	FSB	11	1	00	17DE	FAB5	12:47:29
23546	1415327	250	136	N	LS	632	116	FSB	11	1	08	17DE	96A	12:47:31
23546	1415395	250	129	N	LS	634	116	FSB	11	1	02	17DE	F51F	12:47:33
23546	1415465	250	121	N	LS	634	116	FSB	11	1	00	17DE	DD57	12:47:35
23546	1415530	250	114	N	LS	634	116	FSB	11	1	00	17DE	069C	12:47:37
23546	1415562	250	110	N	LS	634	116	FSB	11	1	08	17DE	6851	12:47:39
23546	1415621	250	103	N	LS	634	116	FSB	11	1	02	17DE	6338	12:47:41
23546	1415703	250	92	N	LS	634	116	FSB	11	1	00	17DE	6338	12:47:41
23546	1415753	250	87	N	LS	634	116	FSB	11	1	08	17DE	A58D	12:47:47
23546	1415777	250	84	N	LS	634	116	FSB	11	1	02	17DE	6FA1	12:47:49
23546	1415822	250	77	N	LS	634	116	FSB	11	1	00	17DE	C10D	12:47:51
23546	1415881	250	66	N	LS	634	116	FSB	11	1	00	17DE	10FC	12:47:53
23546	1415910	250	59	N	LS	634	116	FSB	11	1	08	17DE	1456	12:47:55
23546	1415932	250	56	N	LS	636	116	FSB	11	1	02	17DE	85D5	12:47:57
23546	1415964	250	49	N	LS	636	116	FSB	11	1	00	17DE	D342	12:47:59
23546	1415989	250	41	N	LS	636	116	FSB	11	1	00	17DE	D1F4	12:48:01
23546	1416010	250	33	N	LS	636	116	FSB	11	1	08	17DE	AE50	12:48:03
23546	1416026	250	26	N	LS	636	116	FSB	11	1	02	17DE	000F	12:48:05
23546	1416038	250	18	N	LS	636	116	FSB	11	1	00	17DE	0C8C	12:48:07
23546	1416046	250	10	N	LS	636	116	FSB	11	1	00	17DE	4E65	12:48:09
23546	1416049	250	2	N	LS	636	116	FSB	11	1	08	17DE	32D8	12:48:11
23546	1416049	250	0	N	LS	636	116	-	11	1	02	17DE	189B	12:48:13
23546	1416049	250	0	N	LS	636	116	-	11	1	00	17DE	C01A	12:48:15
23546	1416049	250	0	N	LS	636	116	-	11	1	00	17DE	183B	12:48:17
23546	1416049	250	0	N	LS	636	116	-	11	1	08	17DE	C29A	12:48:19

Figure: FSB applied and Loco stopped eventually, preventing SPAD. Loco switched to LS mode

Text	TimeStamp
Manned LC Gate 536 approaching in 72m	12:47:24
Manned LC Gate 536 approaching in 30m	12:47:25
LS Mode Waiting for Radio Comm with Station	12:47:25
Brake Applied	Speed Limit Exceeded 12:47:30
Ack LS Mode - Radio Comm Fail with Station	12:47:31
LS Mode Waiting for Radio Comm with Station	12:47:37
Ack LS Mode - Radio Comm Fail with Station	12:47:43
LS Mode Waiting for Radio Comm with Station	12:47:49
Ack LS Mode - Radio Comm Fail with Station	12:47:54

Figure: LS mode and Ack alerts as per DMI text Log

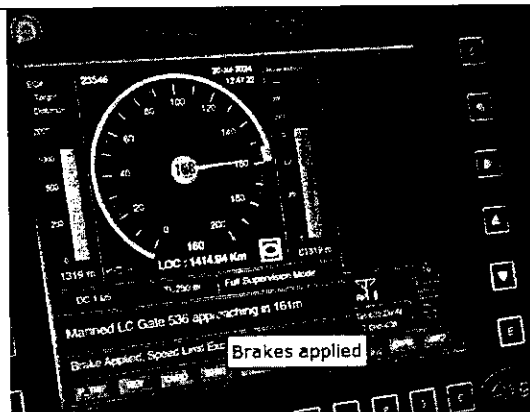


Figure: LP-OCIP showing the application of brakes to prevent SPAD

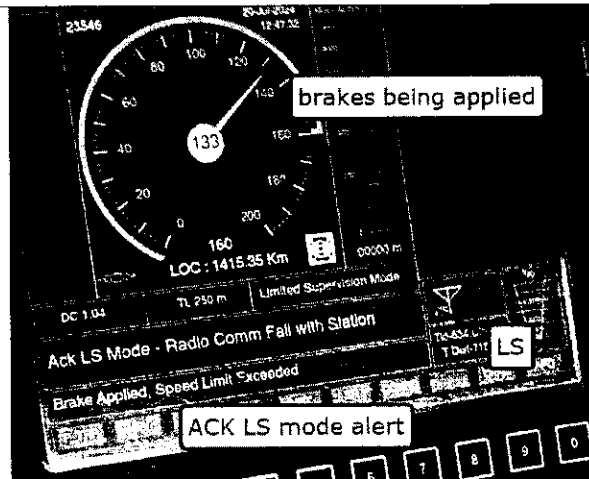


Figure: Loco switched to LS mode due to loss of communication and 'Ack LS mode' alert on LP-OCIP

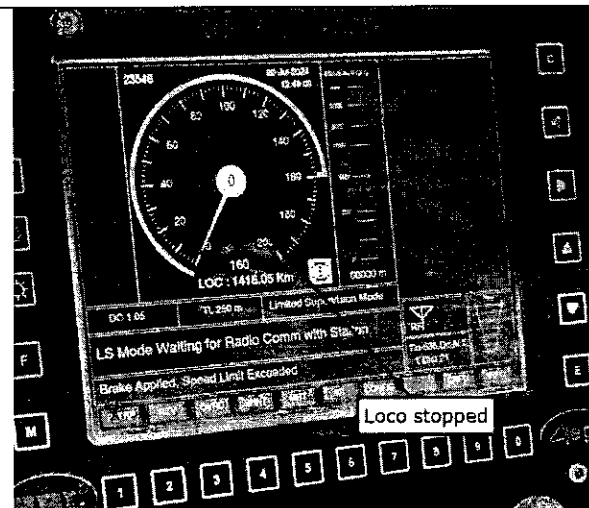


Figure: Loco stopped 274 mtrs before the S2 signal as shown in LP-OCIP

Note: LTCAS retained the MA distance available even after radios were disconnected and continued to apply FSB accordingly to control the speed and prevent SPAD during the mode degradation(FS to LS due to no communication with STCAS in this case).

2.SPAD prevention while radio communication failure in LTCAS (without LS Ack)

Test Scenario	Input Specification	Expected Result	Actual Result
Ensure Loco TCAS is registered with the Rundhi Station TCAS and travelling in FS Mode.	1) Railway staff performs the operation to keep the Home signal S2 aspect as Red 2) Remove the COM ports from the Radio modem-1 and Radio modem-2 in the LTCAS 3) Do not press ACK button	1) Loco TCAS shall maintain FS mode with display of speed, signal aspect and Movement Authority on LP- OCIP 2) HBL personnel shall remove the radio com ports in the LTCAS. 3) Loco shall enter into LS Mode after 10 sec(10 seconds timeout if radios disconnected and communication lost when loco is in the auto section) 4) FSB shall be applied after 7 sec of non-acknowledgment.	1) Loco was travelling at a speed of 160 kmph in DN direction. 2) SPAD prevention test performed at S2 signal. 3) FSB was applied at the Abs Loc 1468.210 Km @160 kmph with an MA of 1363 mtrs. 4) Radio comm ports were disconnected in the LTCAS when Loco was approximately 541 mtrs before the Down auto signal S146721. 5) After 10 sec Loco switched to LS Mode and "ACK LS Mode – Radio Comm Fail with Station" message was displayed on LP – OCIP at Abs Loc 1468.52 Km. 6)The previous FSB(for preventing SPAD by controlling speed before end of authority) continued to apply due to non-acknowledgment of LS mode by LP within 7 secs time window. 7) Loco halted at

Locid	Abs Location	Train Len	Train Speed	Dir	Emergency Mode	Tag-Id	TIN	BrakeStatus	StPrfId	StPrptoMA	HthBits	RL	CRC	Time stamp
23546	1468797	250	160	N	-	LS 190	96	-	9	1	00	64EA 94A1	13:43:01	
23546	1468806	250	160	N	-	LS 190	96	-	9	1	00	64EA 9EE2	13:43:02	
23546	1468815	250	160	N	-	LS 190	96	-	9	1	00	64EA 083E	13:43:05	
23546	1468822	250	160	N	-	LS 190	96	F58	9	1	00	64EA 083E	13:43:06	
23546	1468838	250	158	N	-	LS 192	96	F58	9	1	00	64EA 083E	13:43:09	
23546	1468843	250	150	N	-	LS 192	96	F58	9	1	00	64EA C0C2	13:43:11	
23546	1468850	250	143	N	-	LS 194	96	F58	9	1	00	64EA C0C4	13:43:13	
23546	1468860	250	137	N	-	LS 194	96	F58	9	1	00	64EA 2098	13:43:15	
23546	1468874	250	129	N	-	LS 194	96	F58	9	1	00	64EA 9AA8	13:43:17	
23546	1468791	250	122	N	-	LS 194	96	F58	9	1	00	64EA C961	13:43:19	
23546	1468851	250	116	N	-	LS 194	96	F58	9	1	00	64EA 787A	13:43:21	
23546	1468869	250	111	N	-	LS 194	96	F58	9	1	00	64EA A48A	13:43:23	
23546	1468943	250	105	N	-	LS 194	96	F58	9	1	00	64EA C1D8	13:43:25	
23546	1469026	250	95	N	-	LS 194	96	F58	9	1	00	64EA 00CA	13:43:27	
23546	1469078	250	90	N	-	LS 194	96	F58	9	1	00	64EA 4305	13:43:29	
23546	1469107	250	86	N	-	LS 194	96	F58	9	1	00	64EA 3EB4	13:43:31	
23546	1469149	250	80	N	-	LS 194	96	F58	9	1	00	64EA 1F68	13:43:32	
23546	1469211	250	70	N	-	LS 194	96	F58	9	1	00	64EA 309E	13:43:35	
23546	1469249	250	64	N	-	LS 194	96	F58	9	1	00	64EA 22F8	13:43:37	
23546	1469266	250	60	N	-	LS 194	96	F58	9	1	00	64EA 2AF3	13:43:39	
23546	1469298	250	53	N	-	LS 194	96	F58	9	1	00	64EA 1504	13:43:41	
23546	1469326	250	47	N	-	LS 194	96	F58	9	1	00	64EA 8140	13:43:43	
23546	1469349	250	39	N	-	LS 194	96	F58	9	1	00	64EA 0171	13:43:45	
23546	1469381	250	32	N	-	LS 196	96	F58	9	1	00	64EA E3EE	13:43:47	
23546	1469397	250	24	N	-	LS 196	96	F58	9	1	00	64EA 490A	13:43:49	
23546	1469408	250	16	N	-	LS 196	96	F58	9	1	00	64EA F330	13:43:51	
23546	1469415	250	8	N	-	LS 196	96	F58	9	1	00	64EA C5D2	13:43:53	
23546	1469418	250	0	N	-	LS 196	96	-	9	1	00	64EA 796D	13:43:55	
23546	1469418	250	0	N	-	LS 196	96	-	9	1	00	64EA 918E	13:43:57	
23546	1469418	250	0	N	-	LS 196	96	-	9	1	00	64EA 88AC	13:43:59	
23546	1469418	250	0	N	-	LS 196	96	-	9	1	00	64EA 520C	13:44:01	
23546	1469418	250	0	N	-	LS 196	96	-	9	1	00	64EA EBAD	13:44:03	
23546	1469418	250	0	N	-	LS 196	96	-	9	1	00			

Diagram illustrating the rail yard layout and the sequence of events during the collision:

- Radios disconnected here** (N-96) at 1467.44 (S1(560)DAXT1).
- RDE Comm. Start** (1468.310).
- LC.560 Comm. End** (1468.521).
- Loco stopped here** (R-196, C2T, C2, S2) at 1469.366.
- 250m** distance between RDE Comm. Start and LC.560 Comm. End.
- 210m** distance between Loco stopped here and the end of the yard.
- Timeline of Events:**
 - R-193 (1467.44)
 - S1(560)DAXT1 (1467.44)
 - N-96 (1467.44)
 - R-192 (1468.37)
 - R-194 (1468.52)
 - S1(560) (1468.52)
 - S1(560)DAXT1 (1468.52)
 - S1(560)DAXT1 (1468.52)
 - R-196 (1469.366)
 - R-002 (1469.58)

Figure: RUNDHI station layout

TimeStamp	Locold	FrameOffset	Loco-Sos	TL Ref FNo	TL Ref Offset	TL-Type	Cur-Sig	Cur-Aspct	Nxt-Aspct	MA	AprStnid	AprSigDist	TO Spd	DiHDistTO	TO Rel Dist
13:43:01															
13:43:01	23546	0	0			0	Dn AS	Y			1597	36002	541	0	0
13:44:19															
13:44:21															
13:44:21															
13:44:21															
13:44:23															
13:44:23															
13:44:23															
13:44:25															
13:44:25															
13:44:27	23546	1	0			0	Dn H	R			158	0	158	0	0

Figure: StnRegPkts Log showing no communication packets available when radios were disconnected.

Text	TimeStamp	
Vande Bharat DTC 8 Coaches	13:43:12	
Ack LS Mode - Radio Comm Fail with Station	13:43:12	
LS Mode Waiting for Radio Comm with Station	13:43:18	
Brake Applied	Speed Limit Exceeded	13:43:19
Ack LS Mode - Radio Comm Fail with Station	13:43:23	
LS Mode Waiting for Radio Comm with Station	13:43:29	
Ack LS Mode - Radio Comm Fail with Station	13:43:35	
LS Mode Waiting for Radio Comm with Station	13:43:41	
Ack LS Mode - Radio Comm Fail with Station	13:43:47	
LS Mode Waiting for Radio Comm with Station	13:43:52	

Figure: LS mode and Ack alerts as per DMI text Log

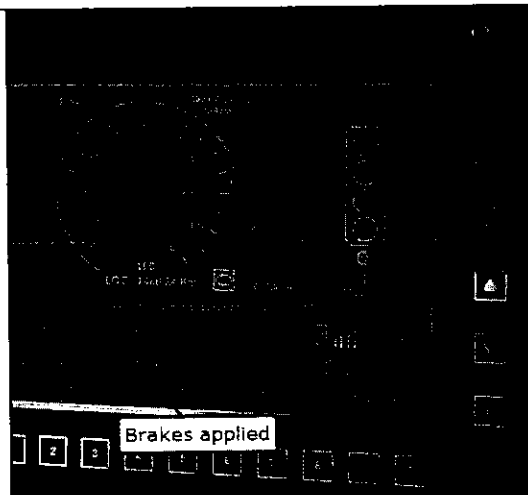


Figure: LP-OCIP showing the application of brakes to prevent SPAD

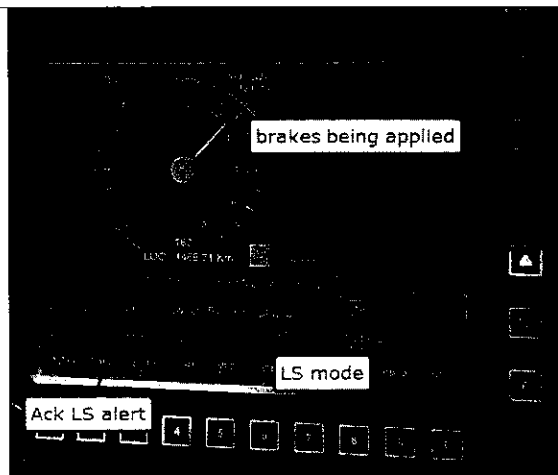


Figure: Loco switched to LS mode due to loss of communication and 'Ack LS mode' alert on LP-OCIP

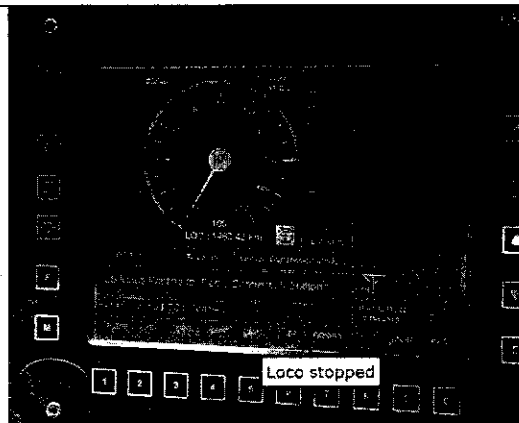


Figure: Loco stopped 158 mtrs before the S2 signal as shown in LP-OCIP

Note: LTCAS retained the MA distance available even after radios were disconnected and continued to apply FSB accordingly to control the speed and prevent SPAD during the mode degradation(FS to LS due to no communication with STCAS).

Lab Trial Report (NCR)

For

Train Collision Avoidance System

Prepared By

HBL Power Systems Limited

Hyderabad

1. Check FS mode to SR mode transition (without SR-Mode Ack) when tags are placed in wrong sequence.

Test Scenario	Input Specification	Expected Result	Actual Result
Ensure Loco TCAS is healthy and registered with the LC-560 TCAS	Railway staff performs the operation at site to route to Main Line. HBL personnel shall swap any two tags on the main line (for which the route is set).	1) HBL personnel informed that the two tags #184 and #186 on the main line to be swapped (Both Main and duplicate tags). 2) After reading Tag – #184, loco shall switch to SR mode. 3) FSB shall be applied by the LTCAS after 7 secs of non-acknowledgment of SR MODE by LP.	1. Loco is travelling at max section speed in DN direction towards Rundhi. 2. After crossing #182, loco read Tag-#186 first and then Tag-#184. 3. After reading #184, direction became un-defined (U) and loco switched to SR Mode and "ACK SR MODE – No section speed info" message was displayed on LP – OCIP. 4. LP did not acknowledge SR mode within the time window of 7 secs, so FSB was applied by the LTCAS (when loco was at speed of 158 kmph). 5. Direction (N here) was defined after reading tag-#190 (N-tag) and loco switched to LS Mode and "Ack LS Mode" message was displayed on DMI. 6. FSB was being applied and the

speed reduced to 100 kmph.

7. After reading second N-Tag #192, Loco switched to FS Mode, FSB was released, and Loco continued to travel.

Locold	Abs Location	Train Len	Train Speed	Dir	Emergency	Mode	Tag-Id	TIN	BrakeStatus	StPrfId	StPrptMA	HlthBits	RL	CRC	Time stamp
30617	1466704	250	158	N	-	FS	182	96	-	9	1	2B	1DFD	FA1A	10:40:03
30617	1466793	250	158	N	-	FS	182	96	-	9	1	02	1DFD	A4CE	10:40:05
30617	1466837	250	158	N	-	FS	182	96	-	9	1	00	1DFD	8078	10:40:07
30617	1466964	250	158	N	-	FS	186	96	-	9	1	10	1DFD	C032	10:40:09
30617	1467052	250	158	N	-	FS	186	96	-	9	1	2B	1DFD	C107	10:40:11
30617	1467229	250	158	N	-	FS	186	96	-	9	1	00	1DFD	78FD	10:40:15
30617	1467317	250	158	N	-	FS	186	96	-	9	1	10	1DFD	0AB2	10:40:17
30617	1467405	250	158	N	-	FS	186	96	-	9	1	2B	1DFD	A8FA	10:40:19
30617	1466710	250	158	U	-	SR Mode	0	0	-	0	0	02	1E73	C16E	10:40:21
30617	0	250	158	U	-	SR Mode	0	0	-	0	0	00	1E73	2E74	10:40:23
30617	0	250	158	U	-	SR Mode	0	0	-	0	0	10	1E73	A395	10:40:25
30617	0	250	158	U	-	SR Mode	0	0	-	0	0	2B	1E73	76DA	10:40:27
30617	0	250	156	U	-	SR Mode	0	0	FSB	0	0	02	1E73	306A	10:40:29
30617	0	250	153	U	-	SR Mode	188	0	FSB	0	0	00	1E73	69E6	10:40:31
30617	0	250	149	U	-	SR Mode	188	0	FSB	0	0	10	1E73	0144	10:40:33
30617	0	250	146	U	-	SR Mode	188	0	FSB	0	0	2B	1E73	8D14	10:40:35
30617	1467440	250	143	N	-	SR Mode	190	96	FSB	0	0	02	1E73	F640	10:40:37
30617	1467504	250	137	N	-	SR Mode	190	96	FSB	0	0	00	1E73	E8EA	10:40:39
30617	1467581	250	137	N	-	SR Mode	190	96	FSB	0	0	10	1E73	B776	10:40:41
30617	1467656	250	133	N	-	LS	190	96	FSB	13	1	2B	1E73	08E0	10:40:43
30617	1467729	250	130	N	-	LS	190	96	FSB	13	1	02	1E73	FF74	10:40:45
30617	1467766	250	129	N	-	LS	190	96	FSB	13	1	00	1E73	14E4	10:40:47
30617	1467836	250	125	N	-	LS	190	96	FSB	13	1	10	1E73	35A7	10:40:49
30617	1467939	250	120	N	-	LS	190	96	FSB	13	1	2B	1E73	CC57	10:40:51

Figure: Mode changed to SR and direction Undefined after tags were read in wrong sequence and then FSB applied, then Loco switched to LS mode. (As per LocoReq pkt log)

Simulation analysis report (NCR)

HBL

Locold	Abs Location	Train Len	Train Speed	Dir	Emergency	Mode	Tag-Id	TIN	BrakeStatus	StPrfId	StPrptoMA	HlthBits	RL	CRC	Time stamp
30617	1468005	250	118	N	-	LS	190	96	FSB	13	1	02	1E73	C2CB	10:40:53
30617	1468038	250	116	N	-	LS	190	96	FSB	13	1	00	1E73	9223	10:40:55
30617	1468102	250	112	N	-	LS	190	96	FSB	13	1	10	1E73	73E9	10:40:57
30617	1468163	250	109	N	-	LS	190	96	FSB	13	1	2B	1E73	936D	10:40:59
30617	1468223	250	106	N	-	LS	190	96	FSB	13	1	02	1E73	B6C0	10:41:01
30617	1468282	250	103	N	-	LS	190	96	FSB	13	1	00	1E73	686F	10:41:03
30617	1468390	250	100	N	-	FS	192	96	FSB	13	1	10	1E73	227A	10:41:05
30617	1468462	250	98	N	-	FS	192	96	-	13	1	2B	1E73	01E5	10:41:07
30617	1468579	250	98	N	-	FS	194	96	-	13	1	02	1E73	3C76	10:41:09
30617	1468634	250	98	N	-	FS	194	96	-	13	1	00	1E73	5682	10:41:11
30617	1468688	250	98	N	-	FS	194	96	-	13	1	2B	1E73	A552	10:41:13
30617	1468743	250	98	N	-	FS	194	96	-	13	1	02	1E73	38D0	10:41:15
30617	1468771	250	98	N	-	FS	194	96	-	13	1	00	1E73	5644	10:41:17
30617	1468826	250	98	N	-	FS	194	96	-	13	1	00	1E73	56D7	10:41:19
30617	1468826	250	98	N	-	FS	194	96	-	13	1	10	1E73	F592	10:41:21

Figure: Loco switched to FS mode after reading N-tag#192 and FSB was released.(As per LocoReq pkt log)

Text	TimeStamp	
Manned LC Gate 560 approaching in 122m	10:40:14:410 02/08/2024	
Manned LC Gate 560 approaching in 78m	10:40:15:218 02/08/2024	
Manned LC Gate 560 approaching in 33m	10:40:16:207 02/08/2024	
Ack SR Mode - No Section Speed Info	10:40:22:012 02/08/2024	
Brake Applied	Speed Limit Exceeded	10:40:29:027 02/08/2024
Ack LS Mode	10:40:45:224 02/08/2024	
Vande Bharat DTC 8 Coaches	10:41:06:620 02/08/2024	
End of Authority in 3999m	10:41:11:825 02/08/2024	
End of Authority in 3972m	10:41:12:827 02/08/2024	
End of Authority in 3945m	10:41:13:828 02/08/2024	

Figure: DMI_text Log showing the Ack SR alert and Ack LS mode alert

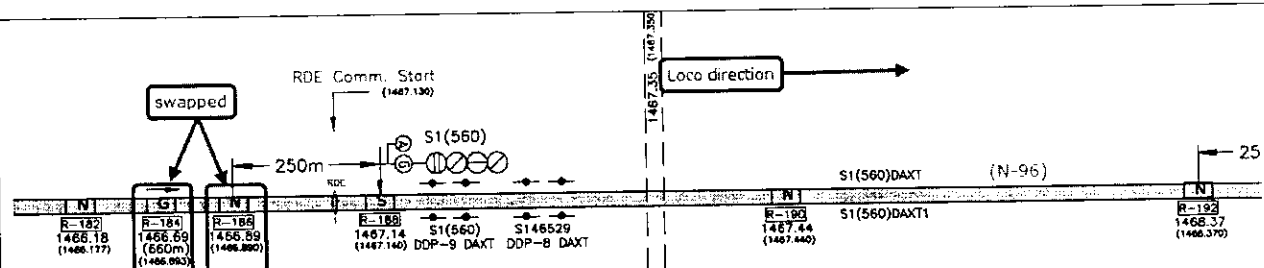


Figure: LC-560 Layout showing the position of swapped tags in the field.

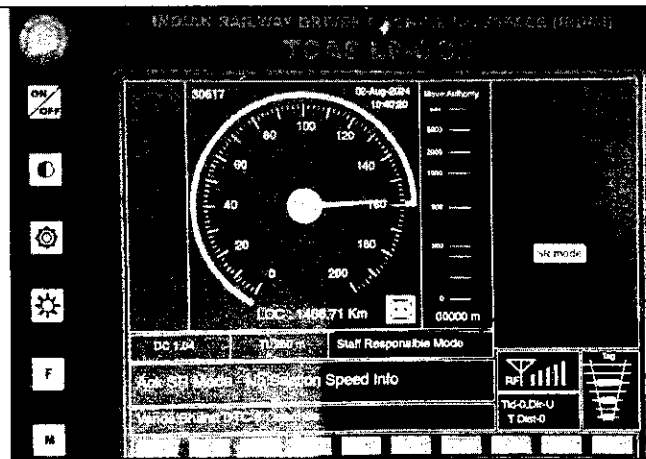


Figure: LP-OCIP showing that Loco switched to SR mode after reading tag#184 and direction became undefined, ack SR mode alert was shown on screen.

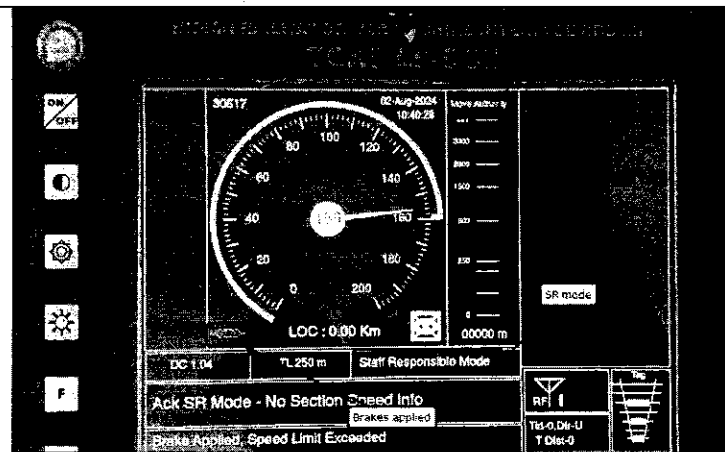


Figure: LP-OCIP showing that brakes were applied after 7 seconds of non-ack of SR mode

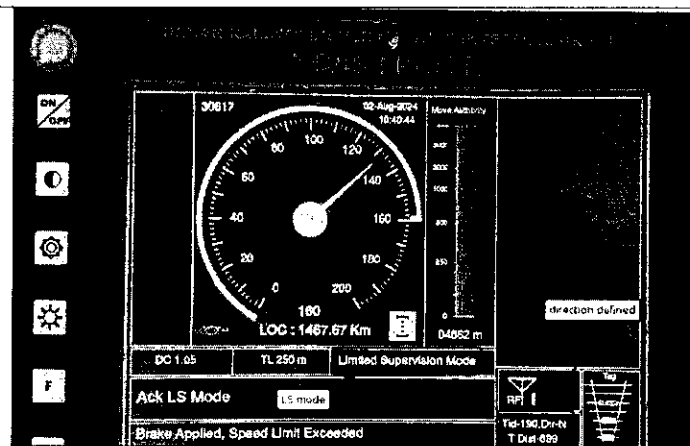


Figure: LP-OCIP showing that Loco switched to LS mode after reading tag#190 and also Loco got the direction

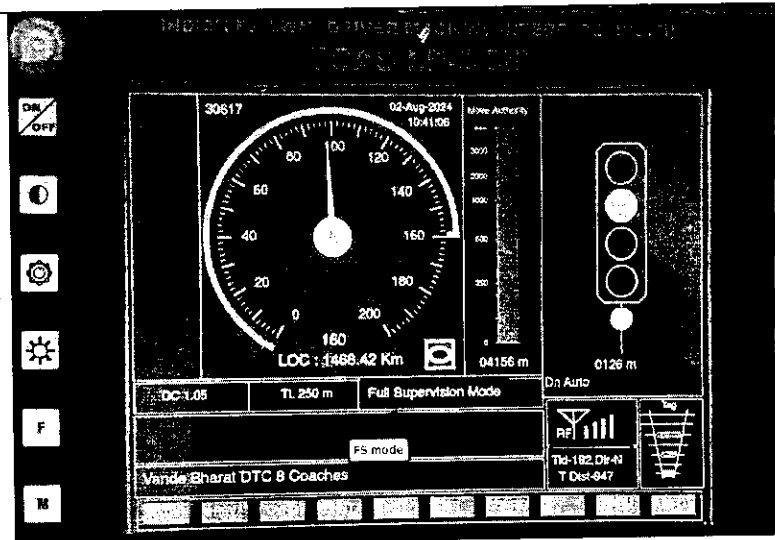


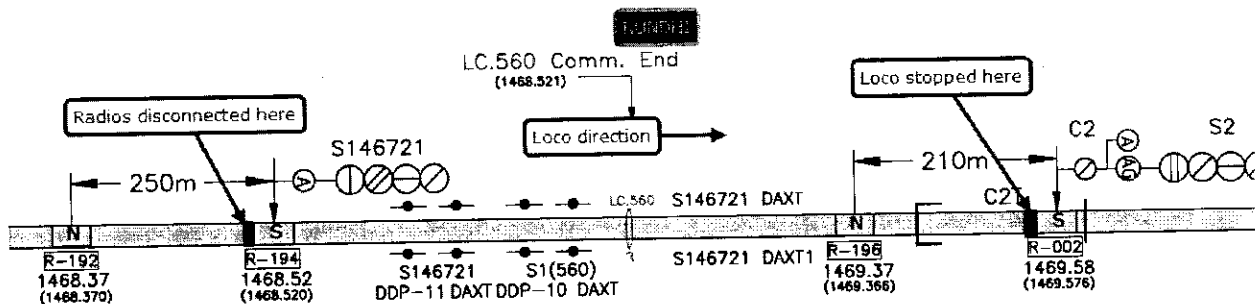
Figure: Loco switched to FS mode after reading second N-tag#192.

2. SPAD prevention while radio communication failure from LTCAS (without LS Ack)

Test Scenario	Input Specification	Expected Result	Actual Result
Ensure Loco TCAS is registered with the Rundhi Station TCAS and travelling in FS Mode.	1) Railway staff performs the operation to keep the Home signal S2 aspect as Red 2) Remove the COM ports from the Radio modem-1 and Radio modem-2 in the LTCAS 3) Do not press ACK button	1) Loco TCAS shall maintain FS mode with display of speed, signal aspect and Movement Authority on LP- OCIP 2) HBL personnel shall remove the radio com ports in the LTCAS. 3) Loco shall enter into LS Mode after 10 sec(10 seconds timeout if radios disconnected and communication lost when loco is in the auto section) 4) FSB shall be applied after 7 sec of non-ack.	1) Loco was travelling at a speed of 160 kmph in Down direction. 2) SPAD prevention test performed at S2 signal. 3) FSB was applied at the Abs Loc 1468.253 @158 Kmph with an MA of 1341 mtrs. 4) Radio comm ports were disconnected in the LTCAS when loco was about 64 mtrs from the Down AutoSignal S146721 5) After 10 sec Loco switched to LS Mode and "ACK LS Mode – Radio Comm Fail with Station" message was displayed on LP – OCIP at Abs Loc 1468.929 Km. 6) The FSB(for preventing SPAD by controlling speed before end of authority) applied due to non-acknowledgment of LS mode by LP within 7 secs time window. 7) Loco halted at Abs Loc 1469.530Km, which is 46 mtrs before the Home signal S2. 8) Loco travelled about 968 mtrs during the

HBL[illegible]

Figure: RUNDHI station layout



Simulation analysis report (NCR)

HBL®

LocoId	Abs Location	Train Len	Train Speed	Dir	Emergency	Mode	Tag-Id	TIN	BrakeStatus	StPrfId	StPrptMA	HthBits	RL	CRC	Time stamp
30617	1468200	250	158	N	-	FS	190	96	-	-	-	02	2B...	EBD8	11:41:41
30617	1468253	250	158	N	-	FS	190	96	FSB	-	-	00	2B...	5976	11:41:43
30617	1468390	250	155	N	-	FS	192	96	FSB	9	1	10	2B...	2EDF	11:41:45
30617	1468464	250	151	N	-	FS	192	96	FSB	9	1	2B	2B...	CB57	11:41:47
30617	1468562	250	144	N	-	FS	194	96	FSB	9	1	02	2B...	87A5	11:41:49
30617	1468640	250	135	N	-	FS	194	96	FSB	9	1	00	2B...	-	11:41:51
30617	1468713	250	126	N	-	FS	194	96	FSB	9	1	2B	2B...	C967	11:41:53
30617	1468819	250	112	N	-	FS	194	96	FSB	9	1	02	2B...	8CF1	11:41:55
30617	1468879	250	103	N	-	FS	194	96	FSB	9	1	00	2CE1	D81D	11:41:57
30617	1468929	250	99	N	-	LS	194	96	FSB	9	1	10	2CE1	A5E8	11:42:01
30617	1468983	250	95	N	-	LS	194	96	FSB	9	1	2B	2CE1	E99B	11:42:03
30617	1469035	250	92	N	-	LS	194	96	FSB	9	1	02	2CE1	5C4C	11:42:05
30617	1469085	250	90	N	-	LS	194	96	FSB	9	1	00	2CE1	84DC	11:42:07
30617	1469163	250	84	N	-	LS	194	96	FSB	9	1	10	2CE1	14DD	11:42:09
30617	1469227	250	79	N	-	LS	194	96	FSB	9	1	2B	2CE1	DC5F	11:42:11
30617	1469270	250	76	N	-	LS	194	96	FSB	9	1	02	2CE1	FFD8	11:42:13
30617	1469295	250	74	N	-	LS	194	96	FSB	9	1	00	2CE1	9342	11:42:15
30617	1469336	250	71	N	-	LS	194	96	FSB	9	1	10	2CE1	87C5	11:42:17
30617	1469390	250	67	N	-	LS	196	96	FSB	9	1	2B	2CE1	89B1	11:42:19
30617	1469417	250	60	N	-	LS	196	96	FSB	9	1	02	2CE1	8E13	11:42:21
30617	1469449	250	51	N	-	LS	196	96	FSB	9	1	00	2CE1	44AD	11:42:23
30617	1469480	250	43	N	-	LS	196	96	FSB	9	1	10	2CE1	0819	11:42:25
30617	1469500	250	33	N	-	LS	196	96	FSB	9	1	2B	2CE1	9947	11:42:27
30617	1469512	250	24	N	-	LS	196	96	FSB	9	1	02	2CE1	2A...	11:42:29
30617	1469523	250	15	N	-	LS	196	96	FSB	9	1	00	2CE1	6145	11:42:31
30617	1469528	250	6	N	-	LS	196	96	FSB	9	1	10	2CE1	45A0	11:42:33
30617	1469530	250	0	N	-	LS	196	96	-	9	1	2B	2CE1	EEED	11:42:35
30617	1469530	250	0	N	-	LS	196	96	-	9	1	02	2CE1	0E5C	11:42:37
30617	1469530	250	0	N	-	LS	196	96	-	9	1	00	2CE1	D6...	11:42:39

Figure: FSB applied and Loco stopped eventually, preventing SPAD. Also, loco switched to LS mode after 10 secs of no communication (as per LocoReq pkt log)

Text	TimeStamp
End of Authority in 702m	11:41:59:315 31/07/2024
Ack LS Mode - Radio Comm Fail with Station	11:41:59:925 31/07/2024
Vande Bharat DTC 8 Coaches	11:41:59:927 31/07/2024
LS Mode Waiting for Radio Comm with Station	11:42:05:752 31/07/2024
Brake Applied	Speed Limit Exceeded 11:42:06:737 31/07/2024
Ack LS Mode - Radio Comm Fail with Station	11:42:11:522 31/07/2024
LS Mode Waiting for Radio Comm with Station	11:42:17:320 31/07/2024
Ack LS Mode - Radio Comm Fail with Station	11:42:23:122 31/07/2024
LS Mode Waiting for Radio Comm with Station	11:42:28:931 31/07/2024
Vande Bharat DTC 8 Coaches	11:42:34:338 31/07/2024
Ack LS Mode - Radio Comm Fail with Station	11:42:34:747 31/07/2024
LS Mode Waiting for Radio Comm with Station	11:42:40:526 31/07/2024

Figure: LS mode and Ack alerts as per DMI text Log



Figure: LP-OCIP showing the application of brakes to prevent SPAD

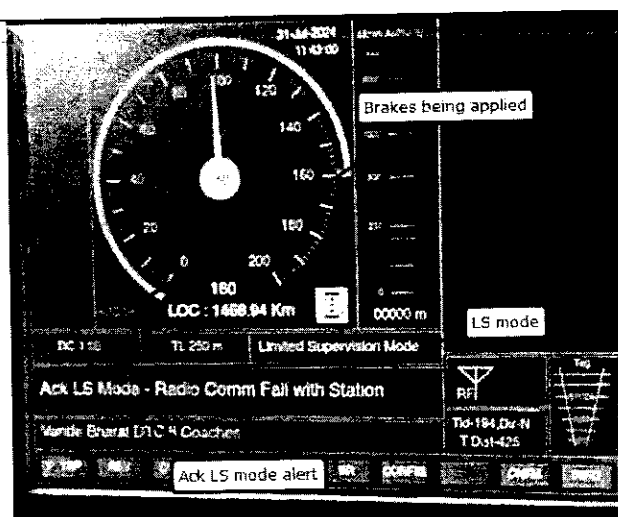


Figure: Loco switched to LS mode due to loss of communication and 'Ack LS mode' alert on LP-OCIP

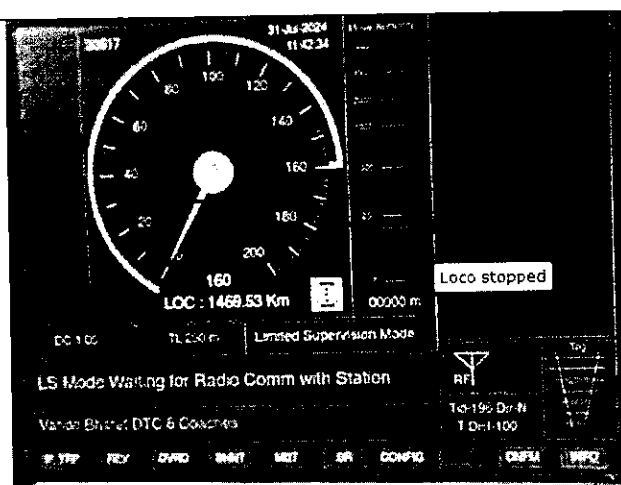


Figure: Loco stopped 46 mtrs before the S2 signal as shown in LP-OCIP

3. SPAD prevention while radio communication failure from LTCAS (without LS Ack)

Test Scenario	Input Specification	Expected Result	Actual Result
Ensure Loco TCAS is registered with the Rundhi Station TCAS and travelling in FS Mode.	1) Railway staff performs the operation to keep the Home signal S2 aspect as Red 2) Remove the COM ports from the Radio modem-1 and Radio modem-2 in the LTCAS 3) Do not press ACK button	1) Loco TCAS shall maintain FS mode with display of speed, signal aspect and Movement Authority on LP- OCIP 2) HBL personnel shall remove the radio com ports in the LTCAS. 3) Loco shall enter into LS Mode after 10 sec(10 seconds timeout if radios disconnected and communication lost when loco is in the auto section) 4) FSB shall be applied after 7 sec of non-ack.	1) Loco was travelling at a speed of 160 kmph in Down direction. 2) SPAD prevention test performed at S2 signal. 3) FSB was applied at the Abs Loc 1468.274 @158 Kmph with an MA of 1302 mtrs. 4)Radio comm ports were disconnected in the LTCAS when loco was about 422 mtrs from the Down AutoSignal S146721 5) After 10 sec Loco switched to LS Mode and "ACK LS Mode – Radio Comm Fail with Station" message was displayed on LP – OCIP at Abs Loc 1468.652 Km. 6) The FSB(for preventing SPAD by controlling speed before end of authority) applied due to non-acknowledgment of LS mode by LP within 7 secs time window. 7) Loco halted at Abs Loc 1469.380Km, which is 196 mtrs before the Home signal S2. 8) Loco travelled about 1176 mtrs during the communication

failure with the station until the loco stopped.

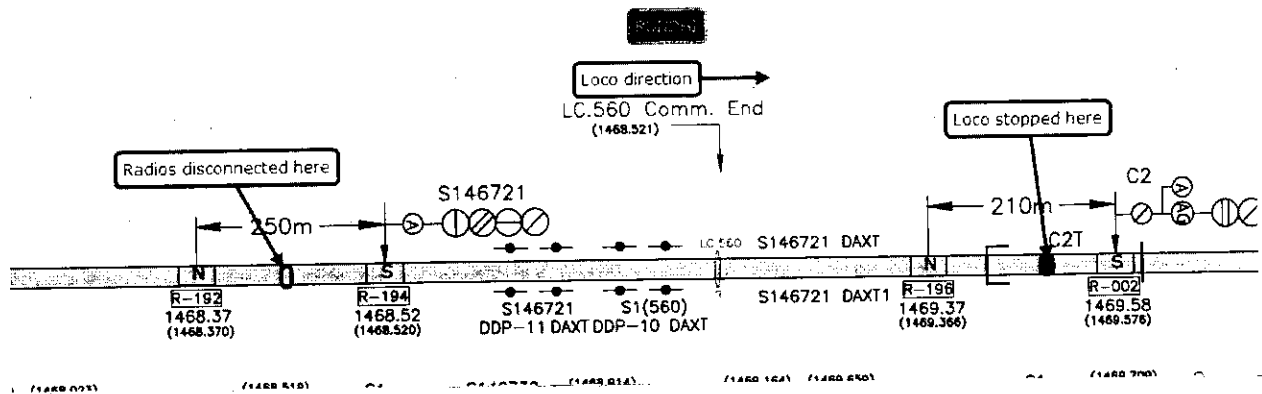


Figure: RUNDHI station layout

Locold	Abs Location	Train Len	Train Speed	Dir	Emergency	Mode	Tag-Id	TIN	BrakeStatus	StPrfld	StPrptoMA	HkHBits	RL	CRC	Time stamp
30617	1467941	250	157	N	-	FS	190	96	-	13	1	2B	3B41	FF51	12:52:27
30617	1468028	250	157	N	-	FS	190	96	-	13	1	02	3B41	51C4	12:52:29
30617	1468107	250	158	N	-	FS	190	96	-	13	1	00	3B41	C20B	12:52:31
30617	1468204	250	157	N	-	FS	190	96	-	13	1	10	3B41	8CEFF	12:52:33
30617	1468291	250	155	N	-	FS	190	96	FSB	13	1	2B	3B41	4C30	12:52:35
30617	1468390	250	152	N	-	FS	192	96	FSB	13	1	00	3B41	4C30	12:52:39
30617	1468473	250	149	N	-	FS	192	96	FSB	13	1	10	3B41	1A8C	12:52:41
30617	1468580	250	145	N	-	FS	194	96	FSB	13	1	2B	3D7D	A019	12:52:43
30617	1468652	250	143	N	-	LS	194	96	FSB	13	1	02	3D7D	A872	12:52:45
30617	1468730	250	139	N	-	LS	194	96	FSB	13	1	00	3D7D	19D4	12:52:47
30617	1468760	250	138	N	-	LS	194	96	FSB	13	1	10	3D7D	FA0A	12:52:49
30617	1468843	250	120	N	-	LS	194	96	FSB	13	1	2B	3D7D	27FD	12:52:51
30617	1468943	250	111	N	-	LS	194	96	FSB	13	1	02	3D7D	CE63	12:52:53
30617	1469007	250	106	N	-	LS	194	96	FSB	13	1	00	3D7D	75FF	12:52:55
30617	1469043	250	96	N	-	LS	194	96	FSB	13	1	10	3D7D	A26C	12:52:57
30617	1469099	250	88	N	-	LS	194	96	FSB	13	1	2B	3D7D	0305	12:52:59
30617	1469145	250	79	N	-	LS	194	96	FSB	13	1	02	3D7D	926E	12:53:01
30617	1469191	250	70	N	-	LS	194	96	FSB	13	1	00	3D7D	15F7	12:53:03
30617	1469233	250	60	N	-	LS	194	96	FSB	13	1	10	3D7D	CEED	12:53:05
30617	1469269	250	51	N	-	LS	194	96	FSB	13	1	2B	3D7D	00AB	12:53:07
30617	1469300	250	43	N	-	LS	194	96	FSB	13	1	02	3D7D	C87A	12:53:09
30617	1469326	250	33	N	-	LS	194	96	FSB	13	1	00	3D7D	E0EF	12:53:11
30617	1469347	250	24	N	-	LS	194	96	FSB	13	1	10	3D7D	040B	12:53:13
30617	1469362	250	8	N	-	LS	194	96	FSB	13	1	2B	3D7D	096A	12:53:15
30617	1469378	250	0	N	-	LS	194	96	-	13	1	02	3D7D	CD...	12:53:17
30617	1469380	250	0	N	-	LS	194	96	-	13	1	00	3D7D	1543	12:53:19
30617	1469380	250	0	N	-	LS	194	96	-	13	1	10	3D7D	AAE0	12:53:21
30617	1469380	250	0	N	-	LS	194	96	-	13	1	2B	3D7D	7FB1	12:53:23

Figure: FSB applied and Loco stopped eventually, preventing SPAD. Also, loco switched to LS mode after 10 secs of no communication (as per LocoReq pkt log)

Stn-Id	Stn-Ver	Abs Lctn	Gen SOS	LocoCount	Crc	TimeStamp	Locold	FrameOffset	Loco-Sos	TL Ref FNo	TL Ref Offset	TL-Type	Cur-Sig	Cur-Aspct	Nxt-Aspct	MA	AprStnId	AprSigDist
36002	1	1471144	0	1	CD12	12:52:29	30617	1	0	-	-	0	Dn AS	Y	R	1740	0	684
36002	1	1471144	0	1	9964	12:52:31	30617	1	0	-	-	0	Dn AS	Y	R	1653	0	597
36002	1	1471144	0	1	648A	12:52:33	30617	1	0	-	-	0	Dn AS	Y	R	1565	0	509
36002	1	1471144	0	0	A718	12:53:35	30617	1	0	-	-	0	Dn AS	Y	R	1478	0	422
36002	1	1471144	0	0	A538	12:53:37												
36002	1	1471144	0	0	6719	12:53:39												
36002	1	1471144	0	1	908A	12:53:41												

no RF communication

Figure: StnRegPkts Log showing no communication between LTCAS and STCAS when radios were disconnected.

Text	TimeStamp	
End of Authority in 982m	12:52:44:474 02/08/2024	
Ack LS Mode - Radio Comm Fail with Station	12:52:45:483 02/08/2024	
Vande Bharat DTC 8 Coaches	12:52:45:720 02/08/2024	
LS Mode Waiting for Radio Comm with Station	12:52:51:276 02/08/2024	
Brake Applied	Speed Limit Exceeded	12:52:52:494 02/08/2024
Ack LS Mode - Radio Comm Fail with Station	12:52:57:086 02/08/2024	
LS Mode Waiting for Radio Comm with Station	12:53:02:877 02/08/2024	
Ack LS Mode - Radio Comm Fail with Station	12:53:08:685 02/08/2024	
LS Mode Waiting for Radio Comm with Station	12:53:14:481 02/08/2024	
Vande Bharat DTC 8 Coaches	12:53:20:094 02/08/2024	
Ack LS Mode - Radio Comm Fail with Station	12:53:20:285 02/08/2024	

Figure: LS mode and Ack alerts as per DMI text Log

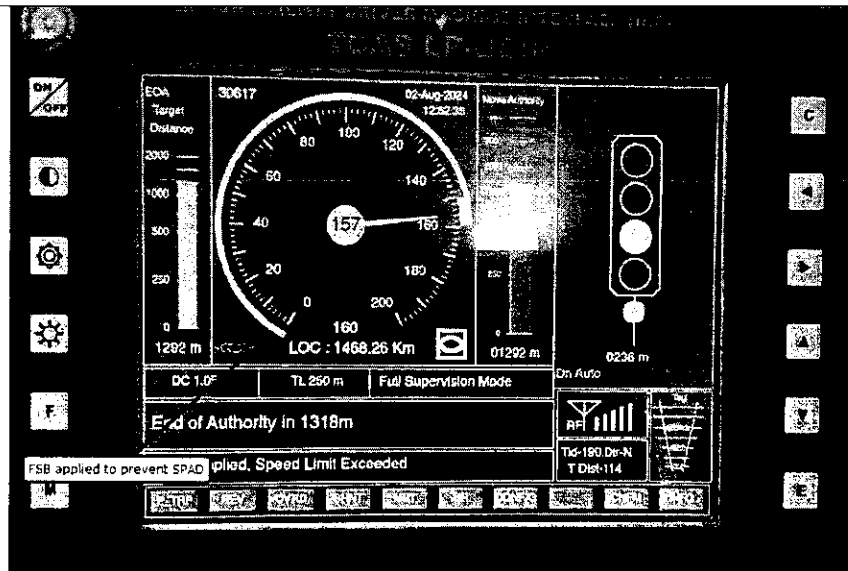


Figure: LP-OCIP showing the application of brakes to prevent SPAD

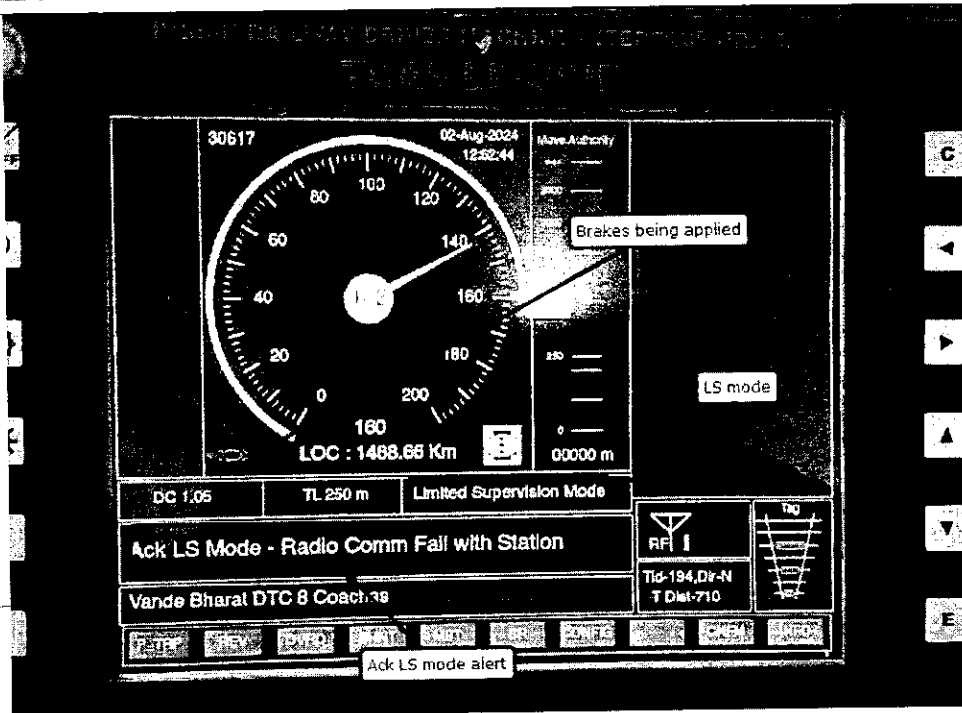


Figure: Loco switched to LS mode due to loss of communication and 'Ack LS mode' alert on LP-OCIP

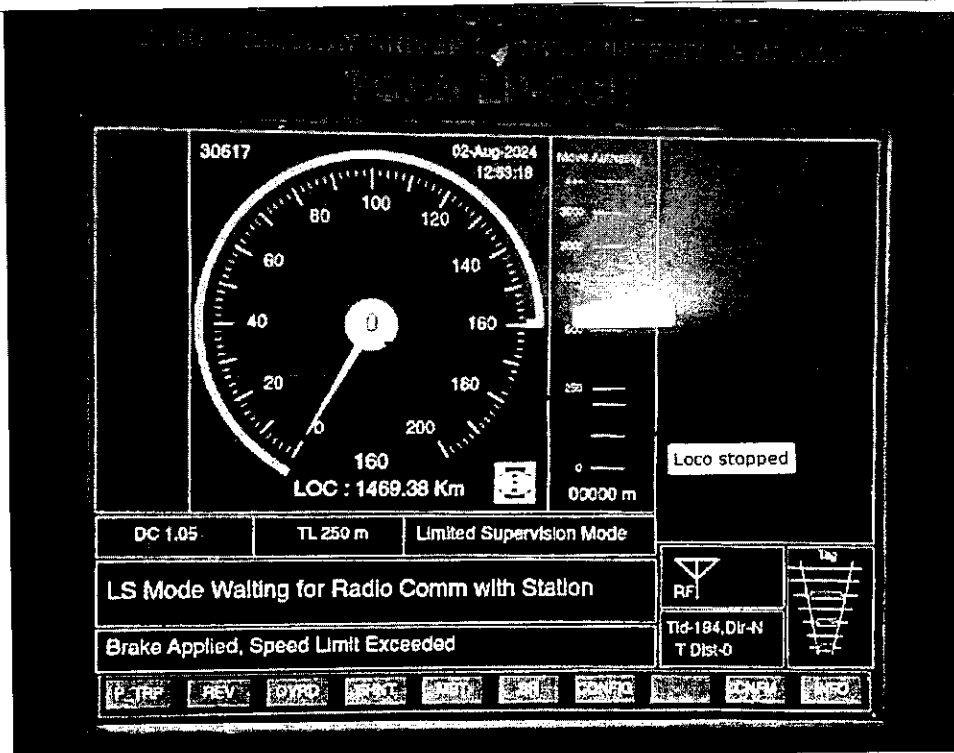


Figure: Loco stopped 196 mtrs before the S2 signal as shown in LP-OCIP